

MICRO X Rev A — printed part reference drawings

Actuated chibi T-rex; 14 policy joints + jaw. Millimetres. Assembly frame: X forward, Y left, Z up, floor Z=0.
20 printed parts (426.6 g solid PLA equivalent) + 18 purchased items (404.0 g catalogue).
Overall views generated from the actual meshes; use STEP for exact geometry. Not production GD&T; prototype fit and fastener checks are still required.

Printed parts

left_yaw2roll	link left_yaw2roll	30.0 x 26.0 x 36.0 mm	5.7 g
left_hip	link left_hip	36.0 x 70.3 x 50.6 mm	18.5 g
left_thigh	link left_thigh	47.8 x 40.2 x 63.1 mm	18.2 g
left_shin	link left_shin	50.7 x 32.7 x 46.1 mm	10.8 g
left_foot	link left_foot	90.0 x 52.0 x 45.2 mm	36.8 g
right_yaw2roll	link right_yaw2roll	30.0 x 26.0 x 36.0 mm	5.7 g
right_hip	link right_hip	36.0 x 70.3 x 50.6 mm	18.5 g
right_thigh	link right_thigh	47.8 x 40.2 x 63.1 mm	18.2 g
right_shin	link right_shin	50.7 x 32.7 x 46.1 mm	10.8 g
right_foot	link right_foot	90.0 x 52.0 x 45.2 mm	36.8 g
chassis	link trunk	138.0 x 58.4 x 64.0 mm	41.9 g
torso_shell	link trunk	110.8 x 84.0 x 60.1 mm	32.6 g
tail_cover	link trunk	78.0 x 66.0 x 48.0 mm	32.7 g
neck_link	link neck	40.1 x 35.5 x 72.1 mm	14.1 g
neck_sleeve	link neck	39.0 x 43.0 x 36.7 mm	12.6 g
head_base	link head_base	24.0 x 30.5 x 31.7 mm	3.6 g
head_yoke	link head_yoke	55.0 x 41.5 x 31.5 mm	20.9 g
head_frame	link head	91.5 x 34.0 x 42.0 mm	16.9 g
skull	link head	126.0 x 81.9 x 64.0 mm	57.8 g
jaw_beak	link jaw	36.1 x 58.4 x 29.0 mm	13.6 g

Purchased items (envelopes)

servo_left_hip_yaw	actuator	link trunk	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_left_hip_roll	actuator	link left_yaw2roll	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_left_hip_pitch	actuator	link left_thigh	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_left_knee	actuator	link left_shin	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_left_ankle	actuator	link left_shin	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_neck_pitch	actuator	link trunk	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_head_pitch	actuator	link neck	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_head_yaw	actuator	link head_yoke	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_head_roll	actuator	link head_yoke	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_right_hip_yaw	actuator	link trunk	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_right_hip_roll	actuator	link right_yaw2roll	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_right_hip_pitch	actuator	link right_thigh	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_right_knee	actuator	link right_shin	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_right_ankle	actuator	link right_shin	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_jaw	actuator	link head	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
battery_np_f550	battery	link trunk	100.0 g	NP-F550 class 7.4 V 2600 mAh removable pack; dimensions 70.8 x 38.4 x
compute_board	electronics	link trunk	30.0 g	Vertical carrier for a compute module (50 x 36 usable outline); PCB pl
camera_module_3	camera	link head	4.0 g	Raspberry Pi Camera Module 3 Standard class, 25 x 24 mm board

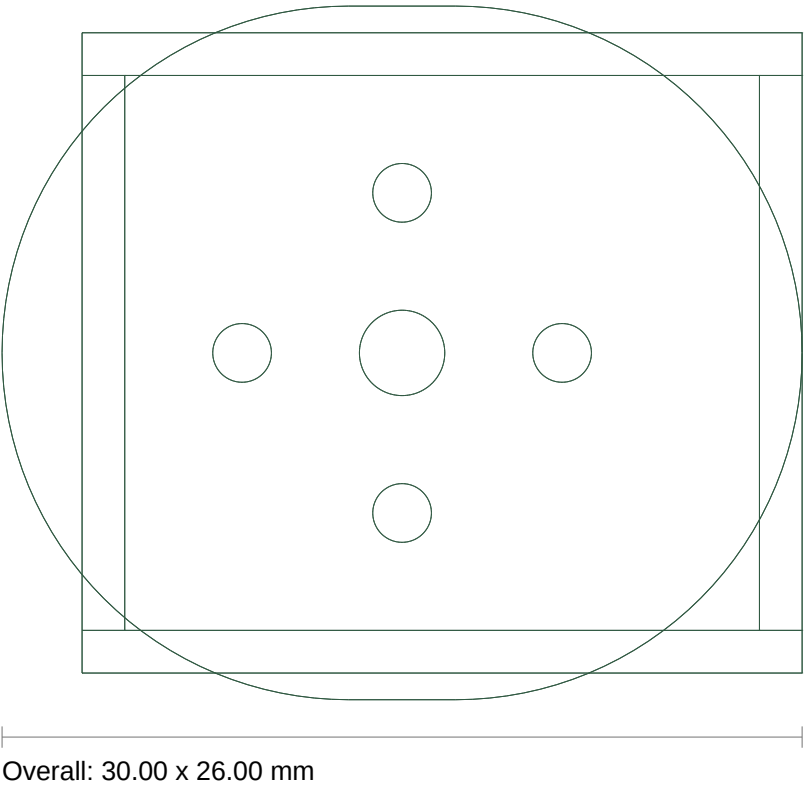
MICRO X Rev A / left_yaw2roll

link: left_yaw2roll | millimetres | original design, prototype reference, not production released

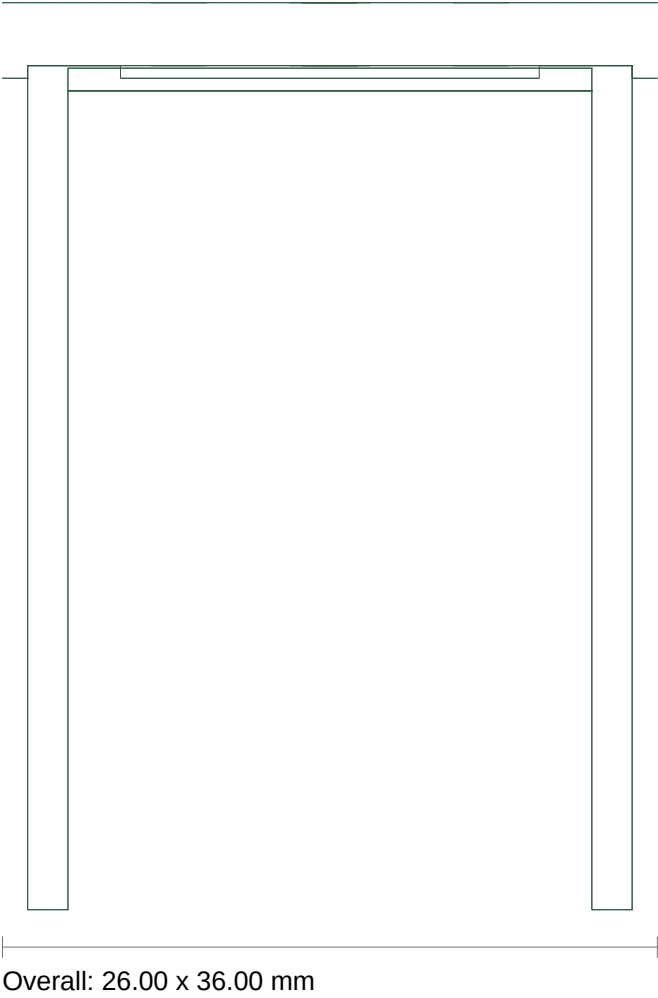
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 30.0 / 26.0 / 36.0 mm

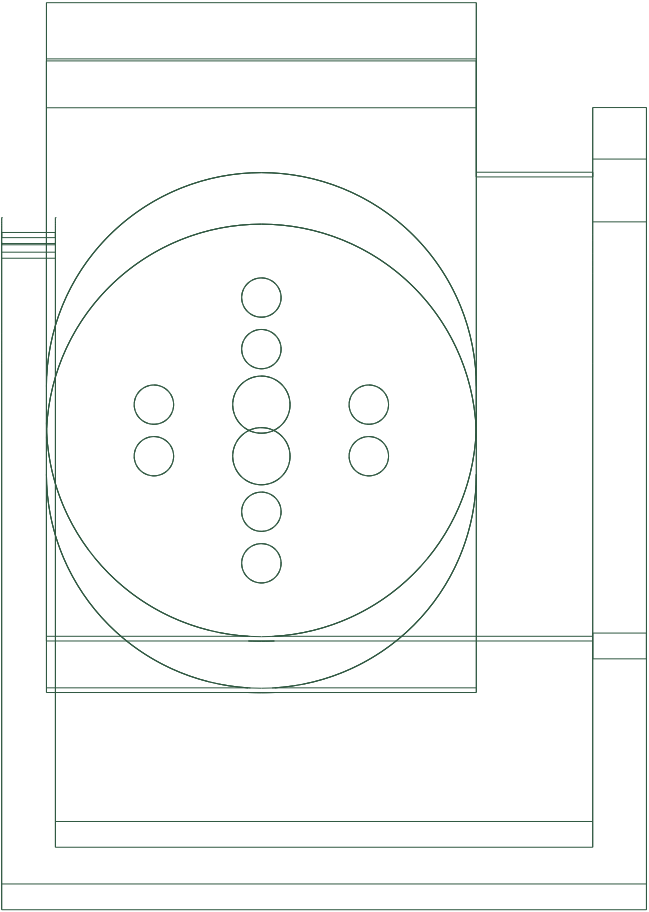
Material volume: 4568.0 mm3 | PLA solid-equivalent: 5.66 g (slicer mass will be lower)

Hip yaw horn plate with hip roll servo cradle; servo drops in from below and is retained by the hip bracket horn/idler bolts. M2 case screws to be confirmed.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / left_hip

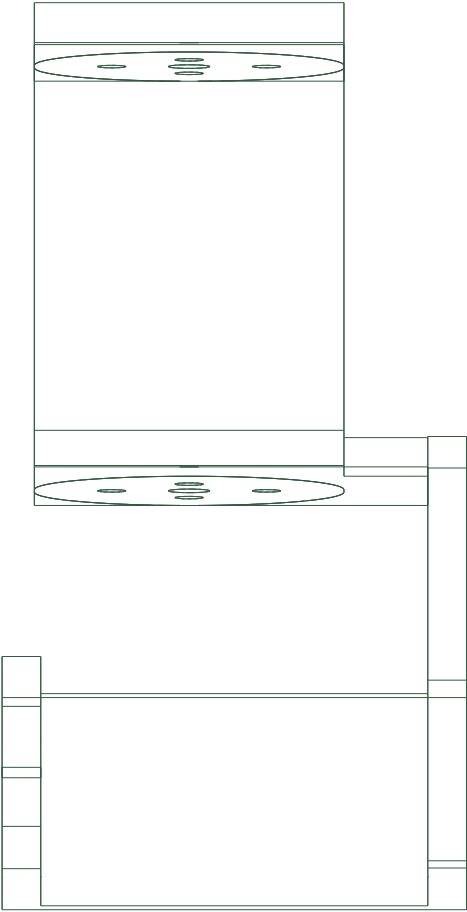
link: left_hip | millimetres | original design, prototype reference, not production released

SIDE X-Z



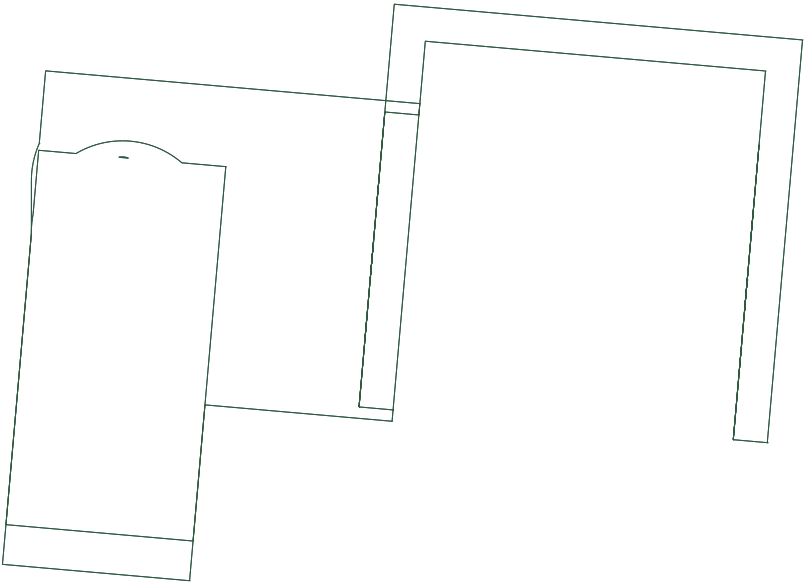
Overall: 36.00 x 50.63 mm

TOP X-Y



Overall: 36.00 x 70.28 mm

FRONT Y-Z



Overall: 70.28 x 50.63 mm

Assembly-frame extents X/Y/Z: 36.0 / 70.28 / 50.63 mm

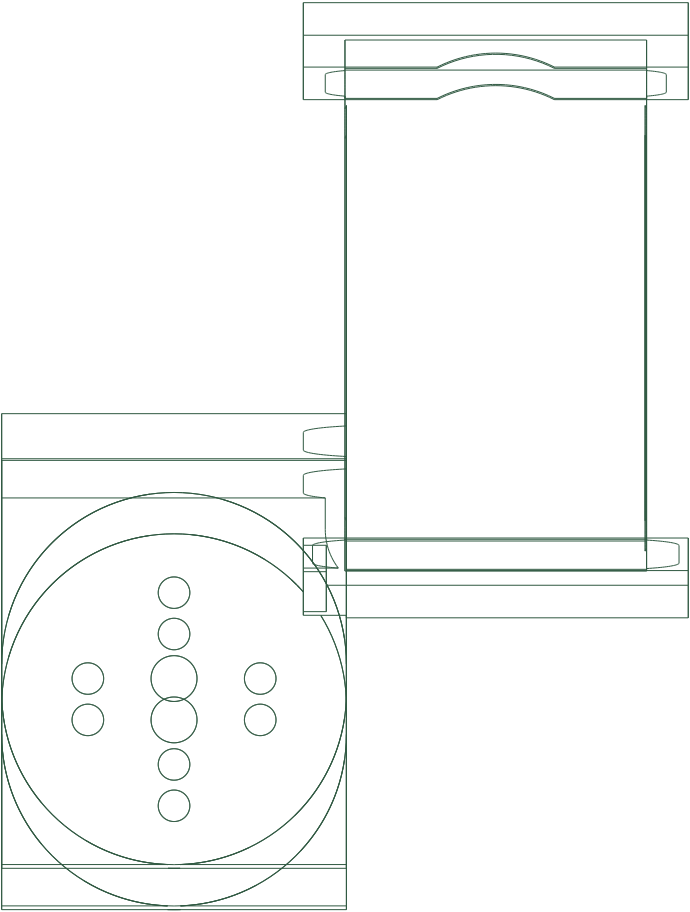
Material volume: 14895.1 mm3 | PLA solid-equivalent: 18.47 g (slicer mass will be lower)

Hip bracket: roll horn/idler plates with an under-bar, pitch horn/idler plates and a tie above the pitch servo cap. Pitch servo body lives in the thigh.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / left_thigh

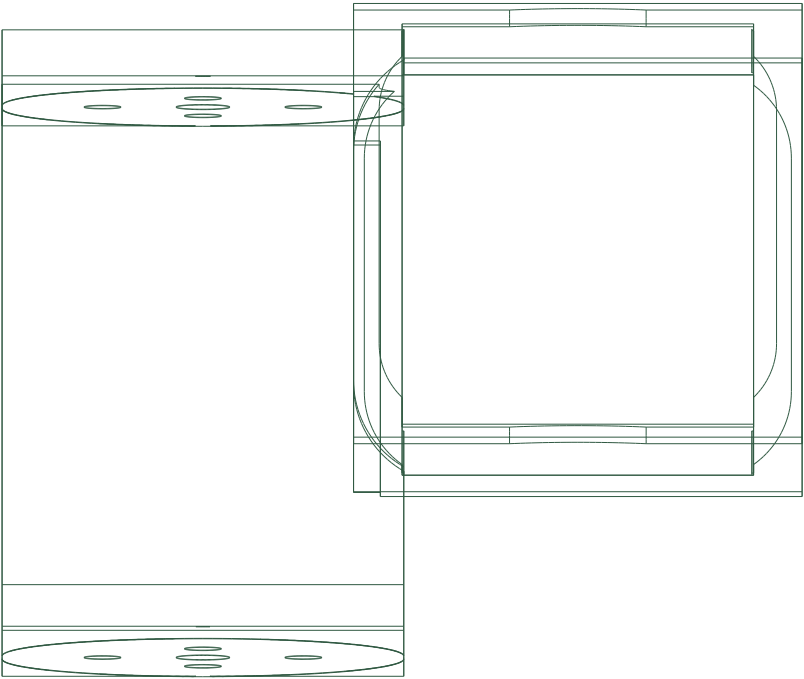
link: left_thigh | millimetres | original design, prototype reference, not production released

SIDE X-Z



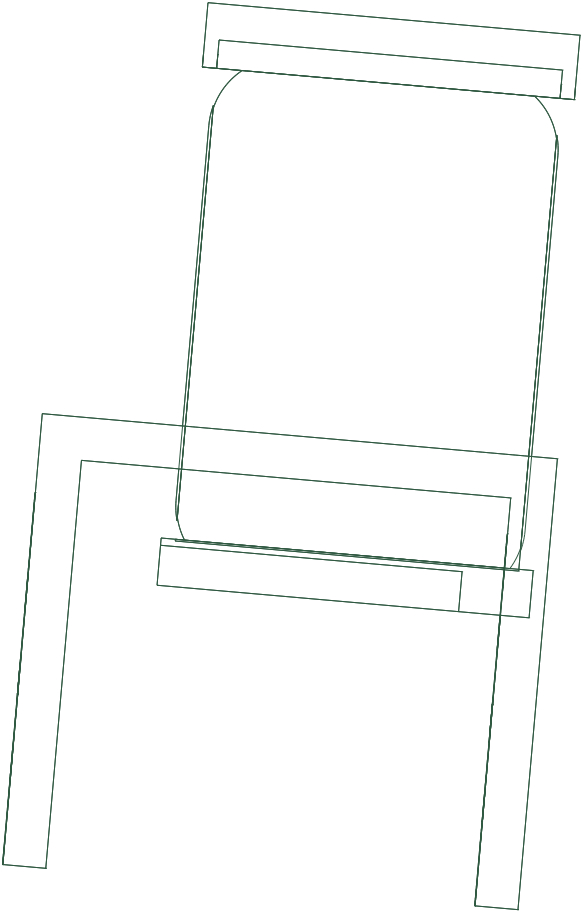
Overall: 47.80 x 63.13 mm

TOP X-Y



Overall: 47.80 x 40.21 mm

FRONT Y-Z



Overall: 40.21 x 63.13 mm

Assembly-frame extents X/Y/Z: 47.8 / 40.21 / 63.13 mm

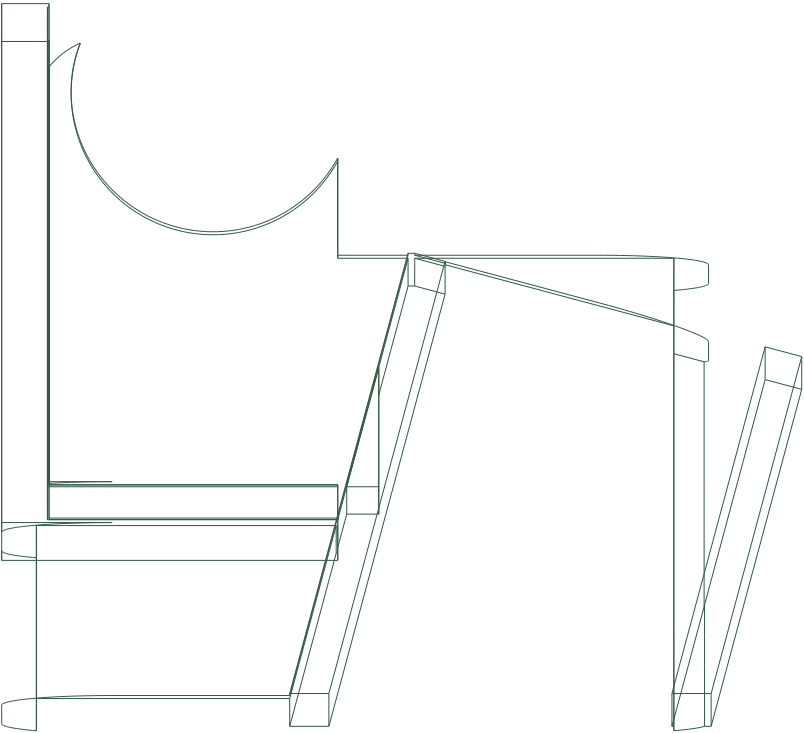
Material volume: 14711.7 mm3 | PLA solid-equivalent: 18.24 g (slicer mass will be lower)

Thigh: houses the hip pitch servo, carries the knee horn and idler plates with a tie above the knee servo.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / left_shin

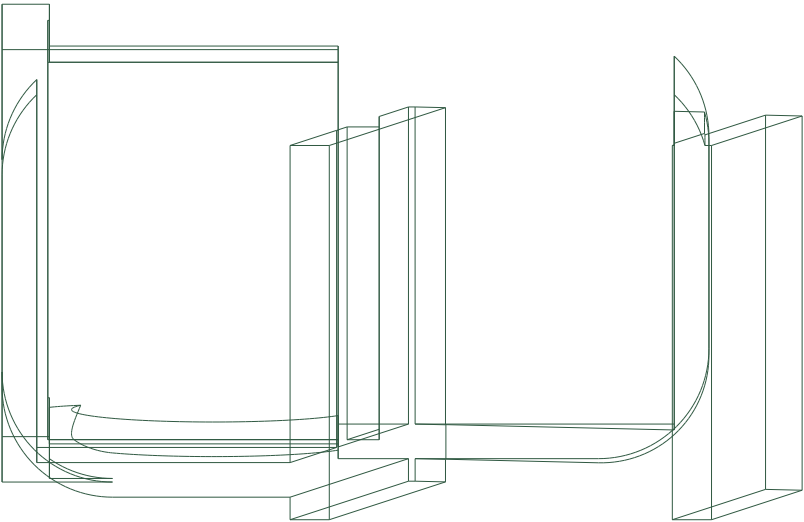
link: left_shin | millimetres | original design, prototype reference, not production released

SIDE X-Z



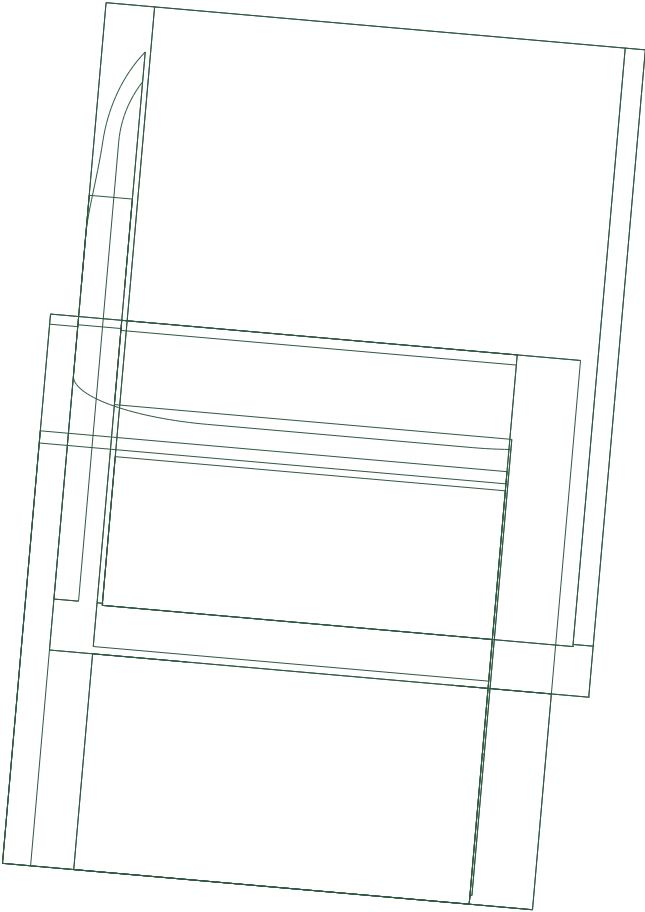
Overall: 50.71 x 46.06 mm

TOP X-Y



Overall: 50.71 x 32.66 mm

FRONT Y-Z



Overall: 32.66 x 46.06 mm

Assembly-frame extents X/Y/Z: 50.71 / 32.66 / 46.06 mm

Material volume: 8706.8 mm3 | PLA solid-equivalent: 10.8 g (slicer mass will be lower)

Shin: spine plate with knee servo bay above an upright ankle servo cradle; both servos retained by the neighbouring clevis plates.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

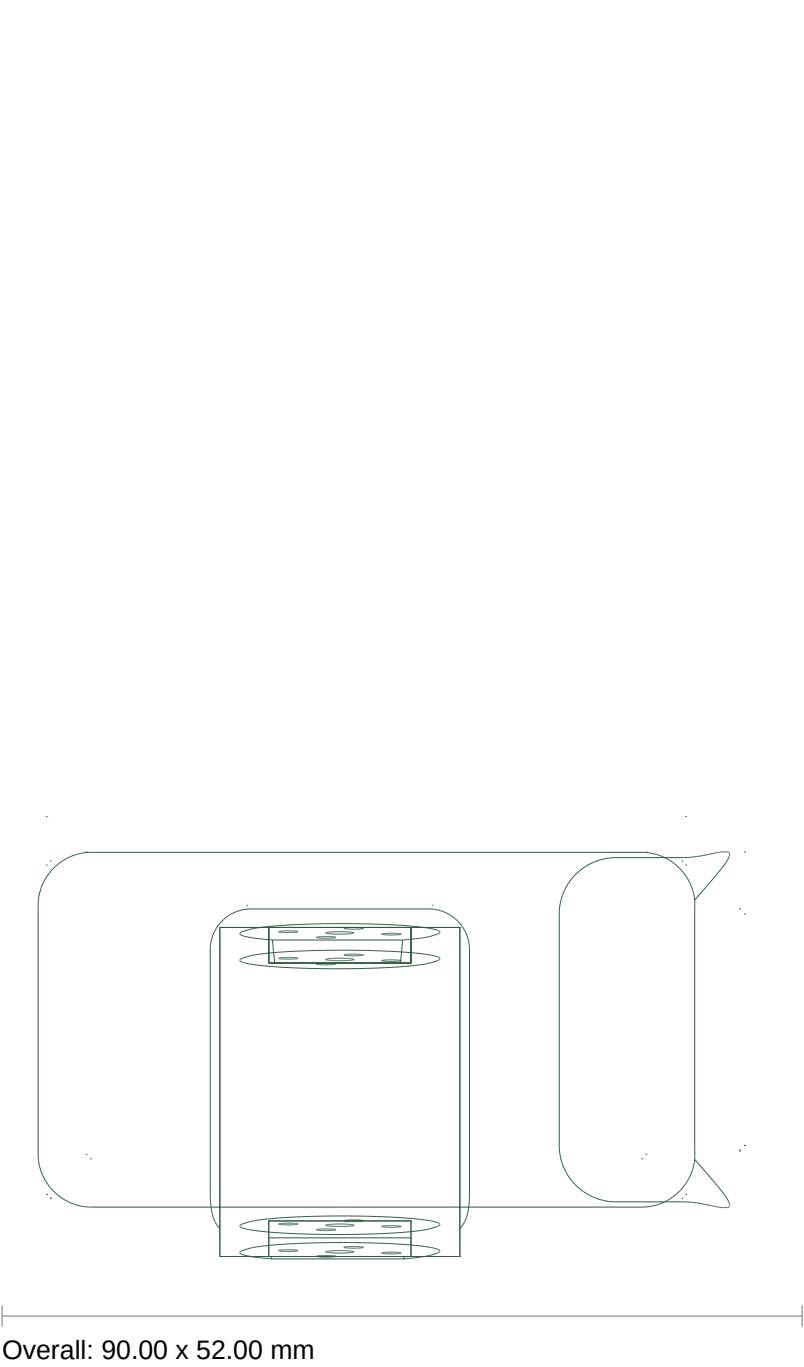
MICRO X Rev A / left_foot

link: left_foot | millimetres | original design, prototype reference, not production released

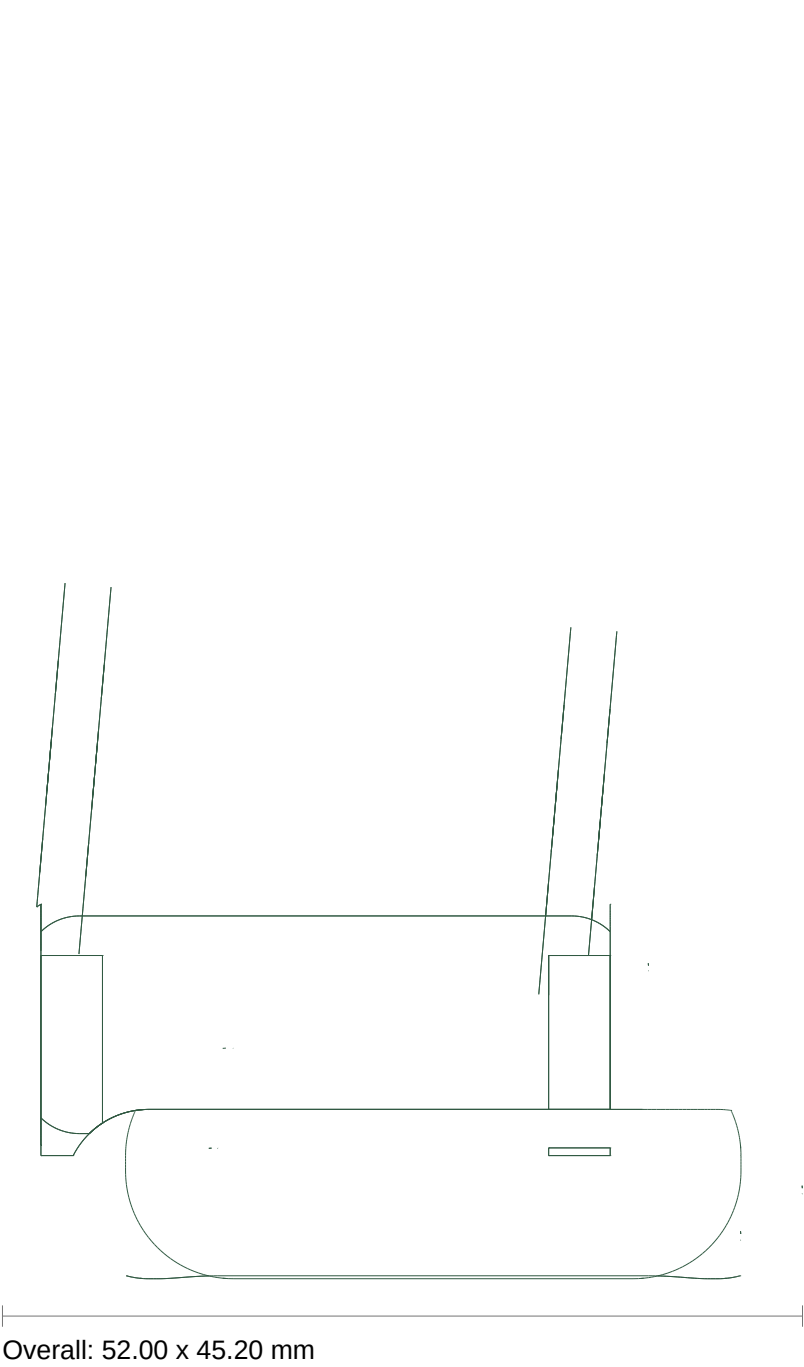
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 90.0 / 52.0 / 45.2 mm

Material volume: 29702.6 mm3 | PLA solid-equivalent: 36.83 g (slicer mass will be lower)

Foot: ankle horn/idler plates on a wide rounded sole with a toe bumper; replaceable TPU sole pad is a follow-up.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

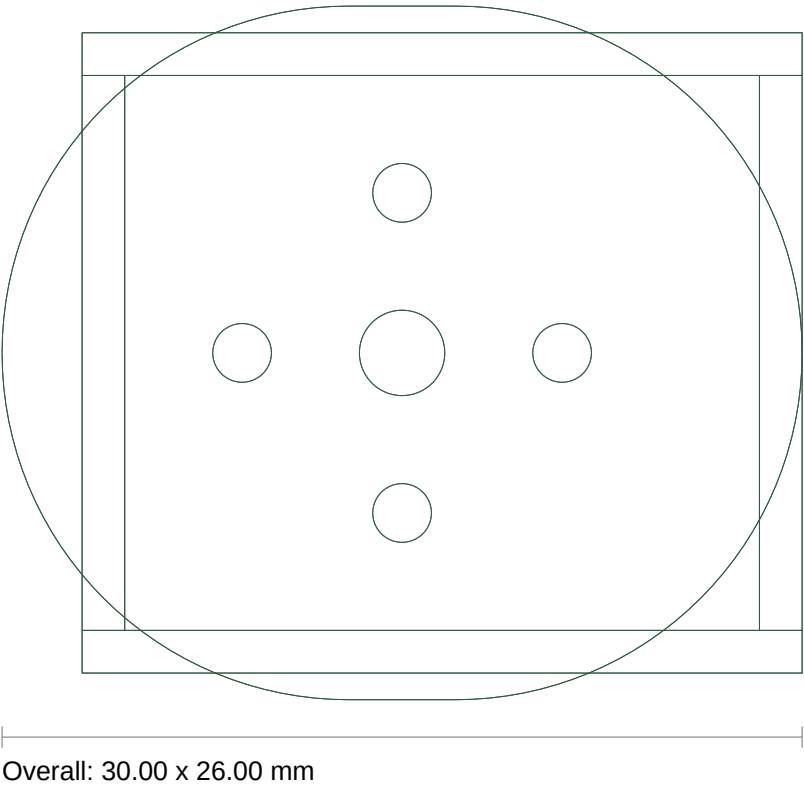
MICRO X Rev A / right_yaw2roll

link: right_yaw2roll | millimetres | original design, prototype reference, not production released

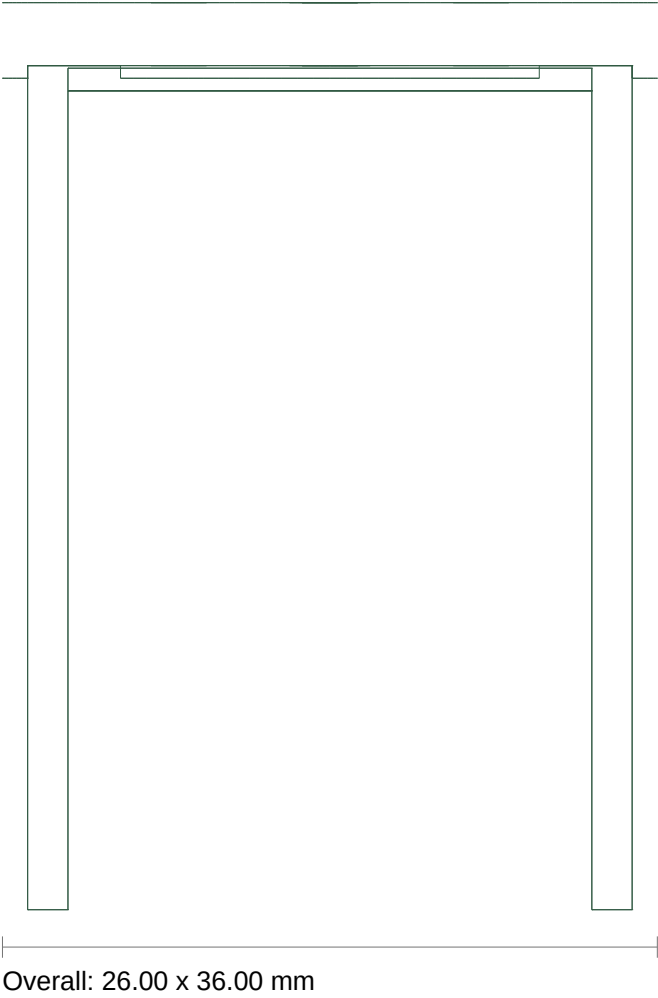
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 30.0 / 26.0 / 36.0 mm

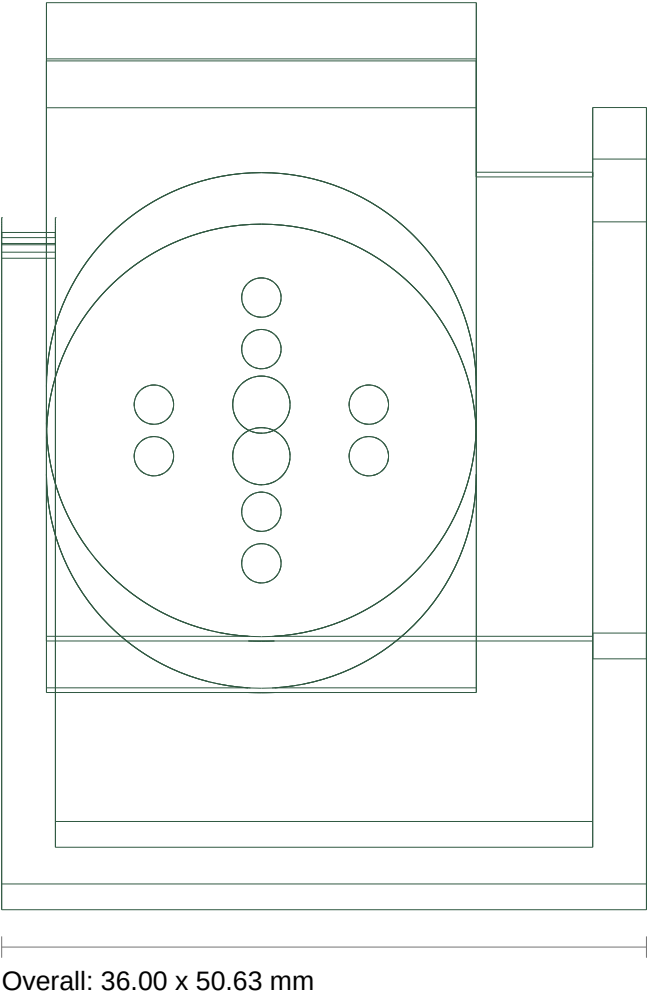
Material volume: 4568.0 mm3 | PLA solid-equivalent: 5.66 g (slicer mass will be lower)

Hip yaw horn plate with hip roll servo cradle; servo drops in from below and is retained by the hip bracket horn/idler bolts. M2 case screws to be confirmed.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

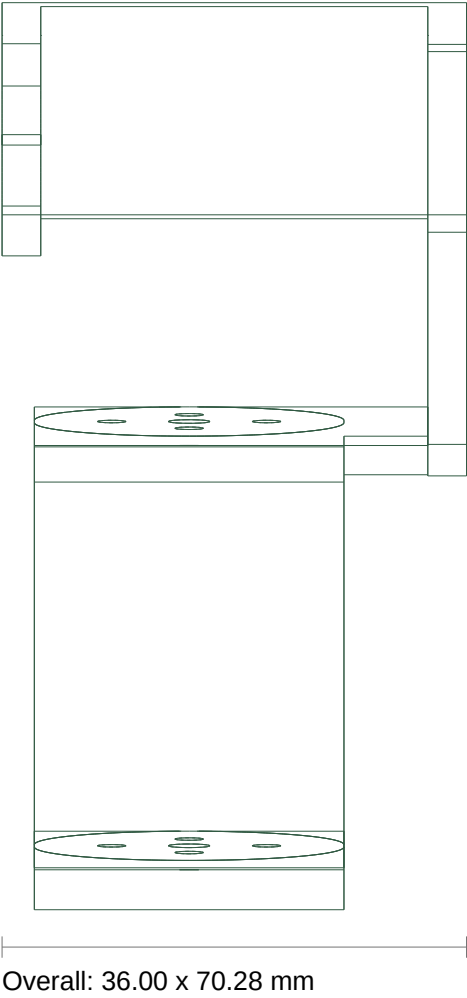
MICRO X Rev A / right_hip

link: right_hip | millimetres | original design, prototype reference, not production released

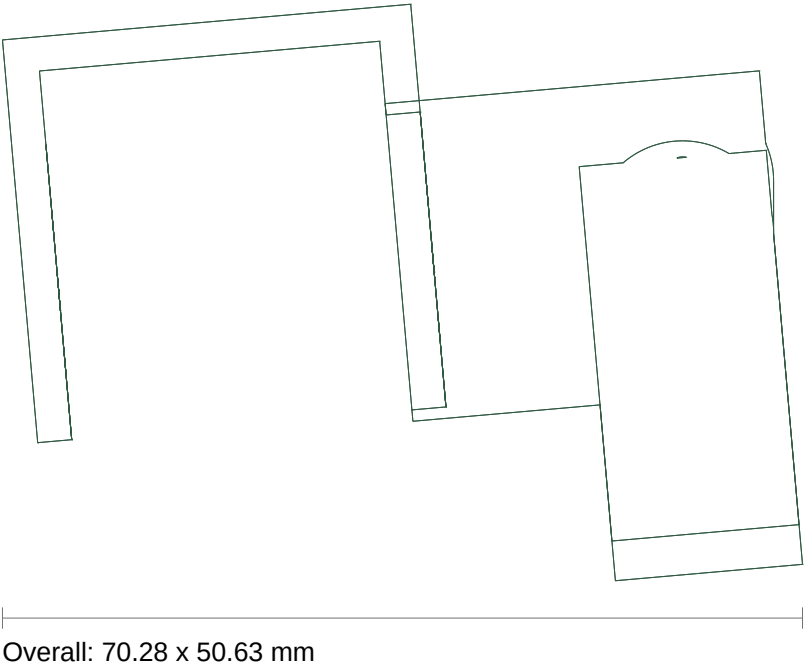
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 36.0 / 70.28 / 50.63 mm

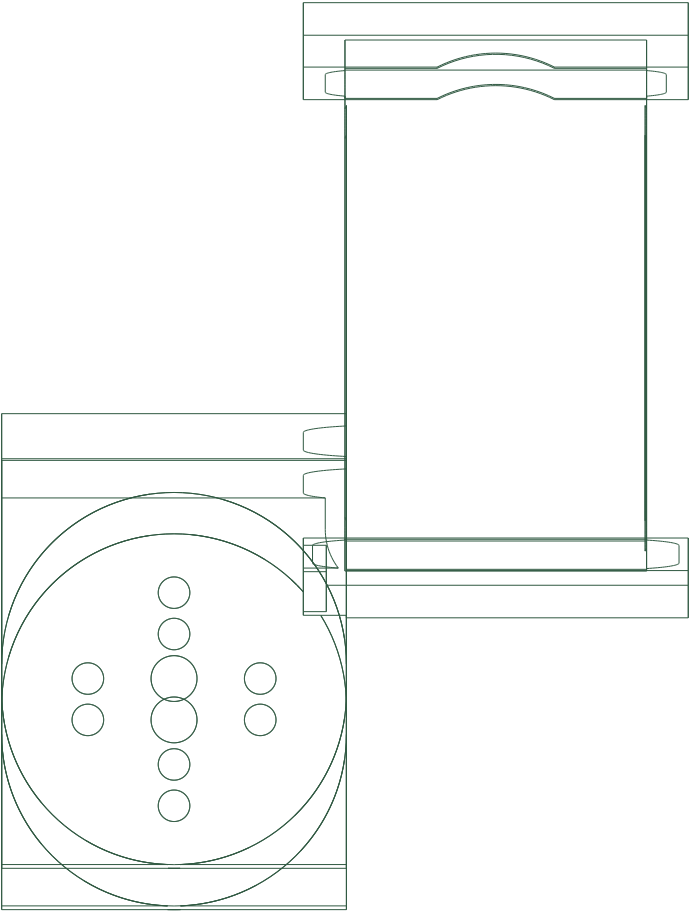
Material volume: 14895.1 mm3 | PLA solid-equivalent: 18.47 g (slicer mass will be lower)

Hip bracket: roll horn/idler plates with an under-bar, pitch horn/idler plates and a tie above the pitch servo cap. Pitch servo body lives in the thigh. Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / right_thigh

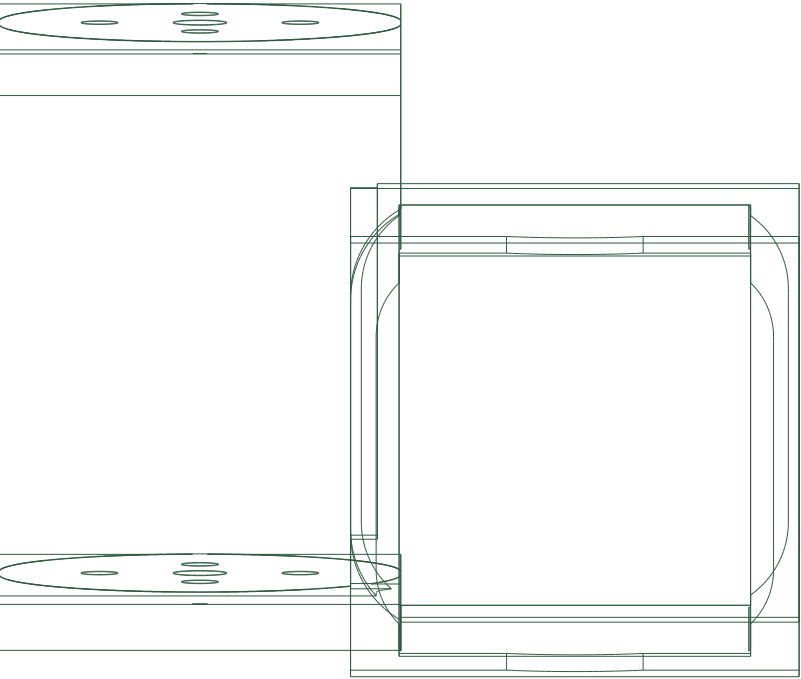
link: right_thigh | millimetres | original design, prototype reference, not production released

SIDE X-Z



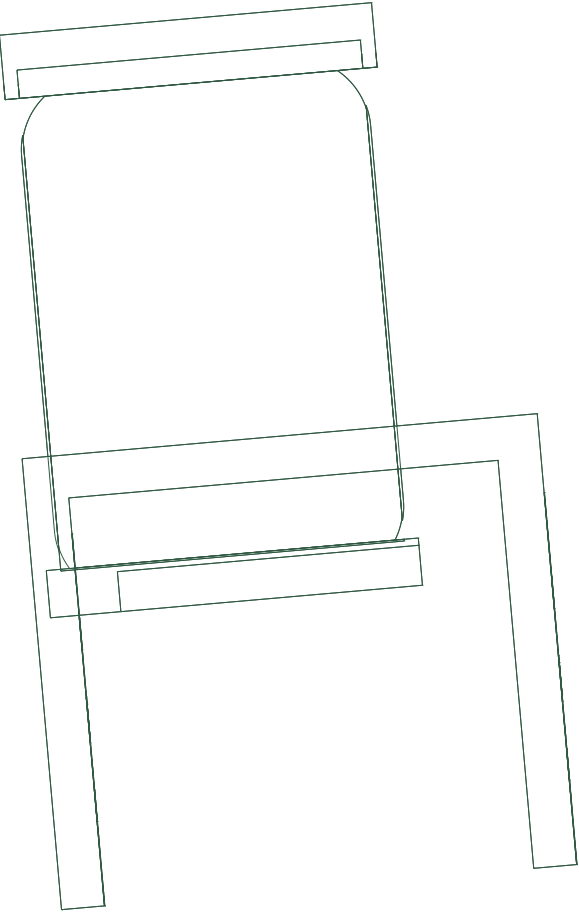
Overall: 47.80 x 63.13 mm

TOP X-Y



Overall: 47.80 x 40.21 mm

FRONT Y-Z



Overall: 40.21 x 63.13 mm

Assembly-frame extents X/Y/Z: 47.8 / 40.21 / 63.13 mm

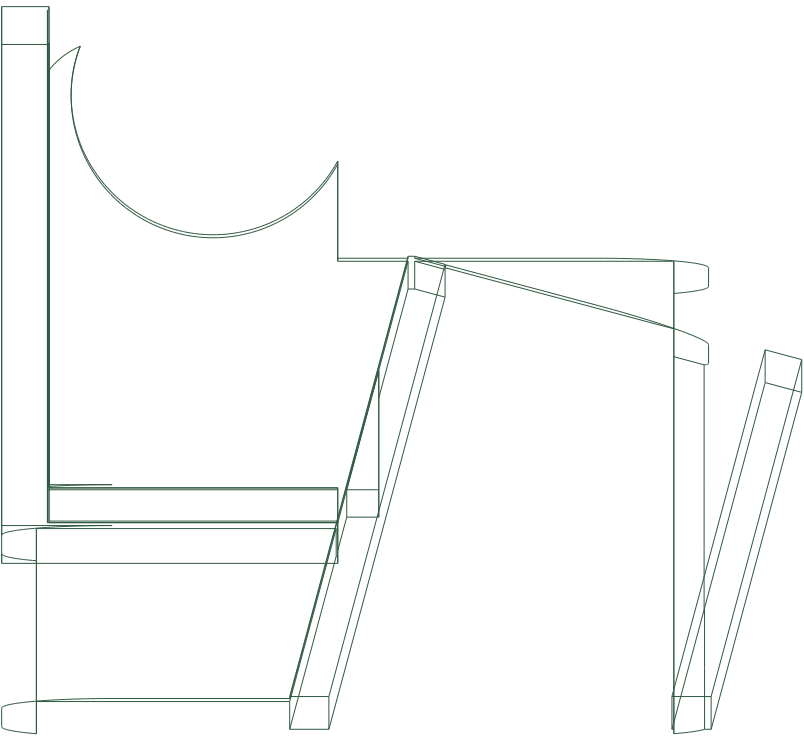
Material volume: 14711.7 mm3 | PLA solid-equivalent: 18.24 g (slicer mass will be lower)

Thigh: houses the hip pitch servo, carries the knee horn and idler plates with a tie above the knee servo.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / right_shin

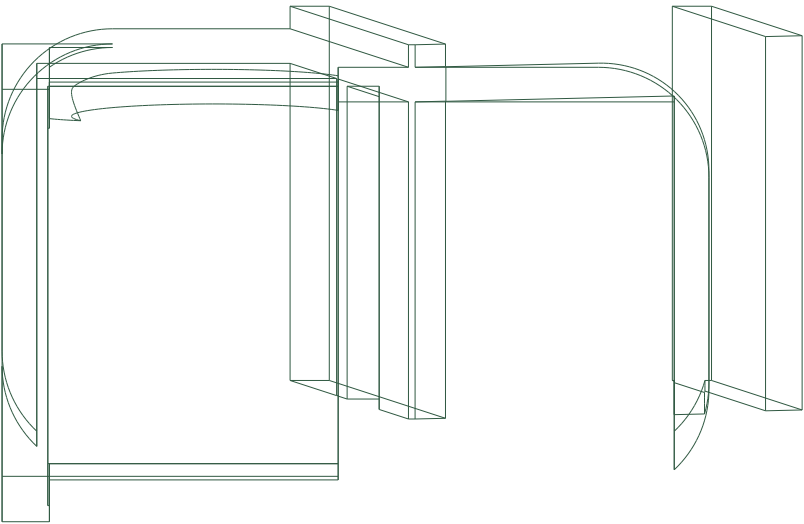
link: right_shin | millimetres | original design, prototype reference, not production released

SIDE X-Z



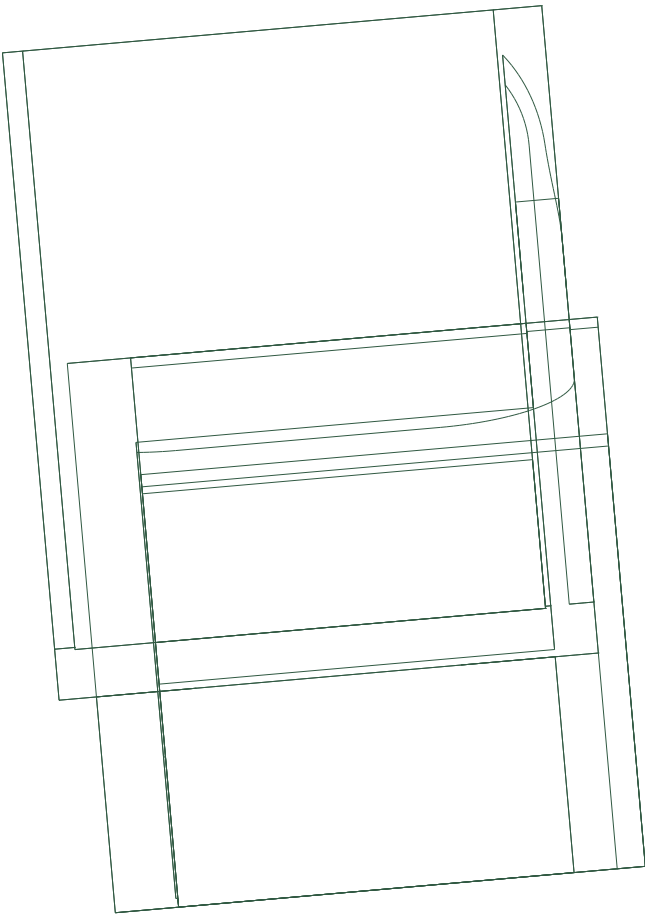
Overall: 50.71 x 46.06 mm

TOP X-Y



Overall: 50.71 x 32.66 mm

FRONT Y-Z



Overall: 32.66 x 46.06 mm

Assembly-frame extents X/Y/Z: 50.71 / 32.66 / 46.06 mm

Material volume: 8706.8 mm3 | PLA solid-equivalent: 10.8 g (slicer mass will be lower)

Shin: spine plate with knee servo bay above an upright ankle servo cradle; both servos retained by the neighbouring clevis plates.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

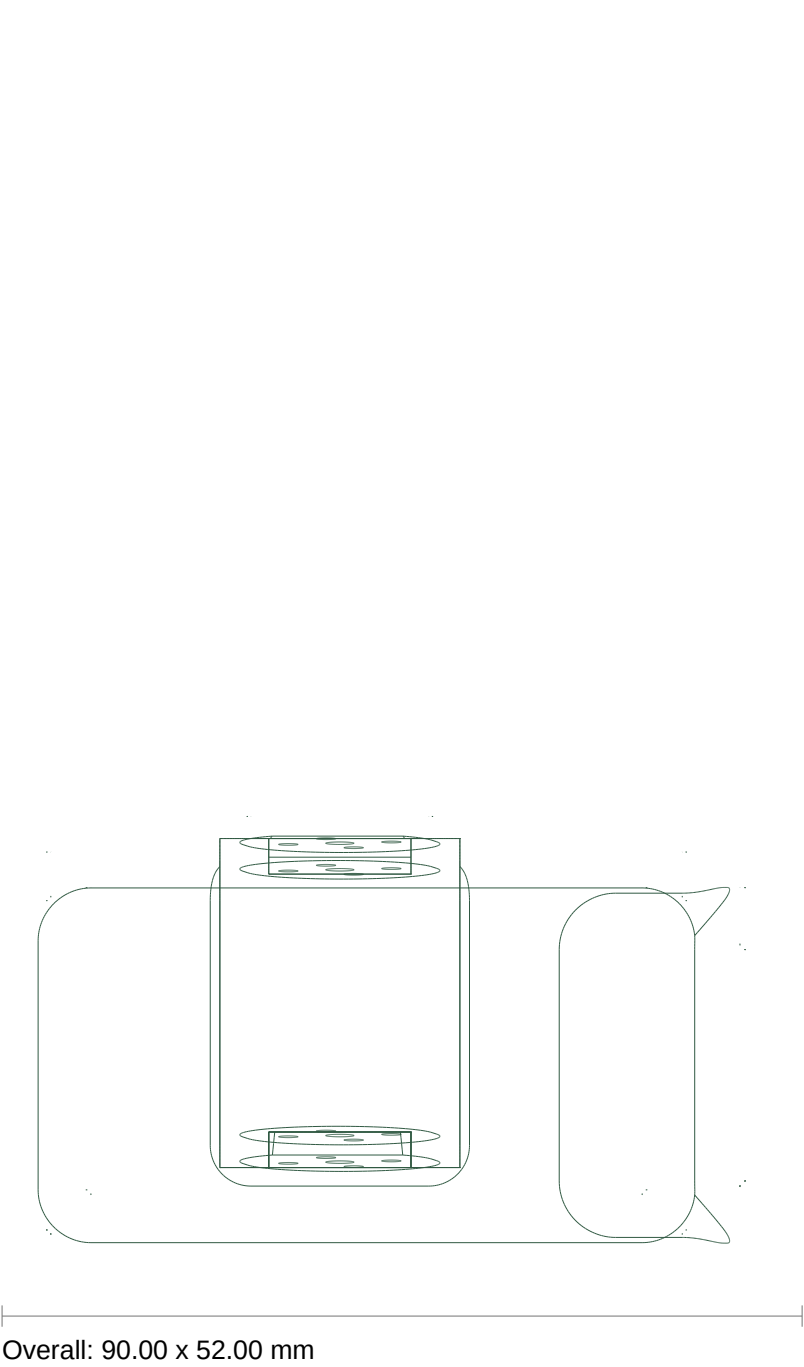
MICRO X Rev A / right_foot

link: right_foot | millimetres | original design, prototype reference, not production released

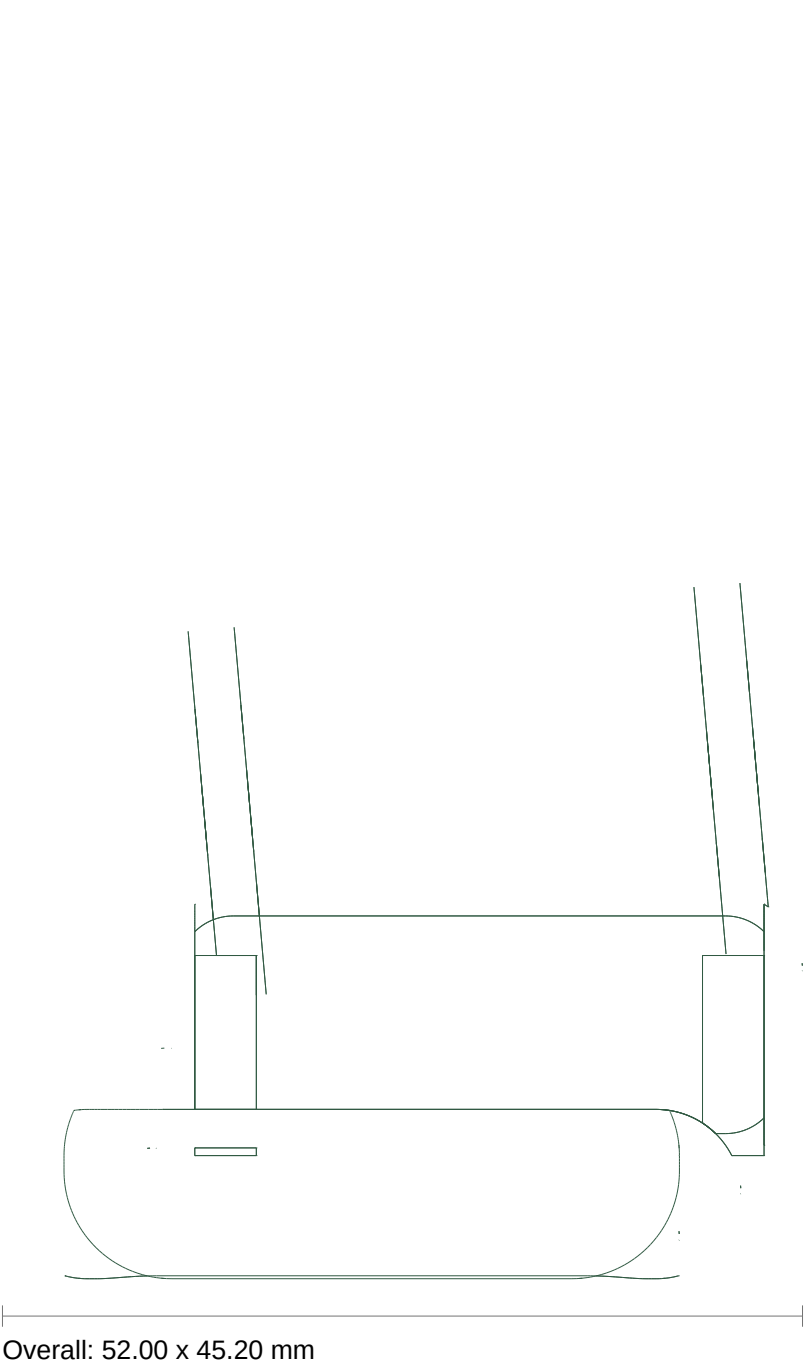
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 90.0 / 52.0 / 45.2 mm

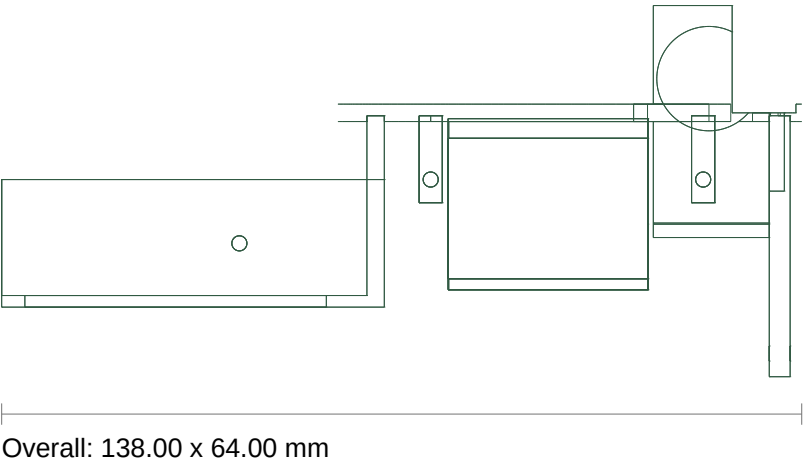
Material volume: 29702.6 mm3 | PLA solid-equivalent: 36.83 g (slicer mass will be lower)

Foot: ankle horn/idler plates on a wide rounded sole with a toe bumper; replaceable TPU sole pad is a follow-up.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

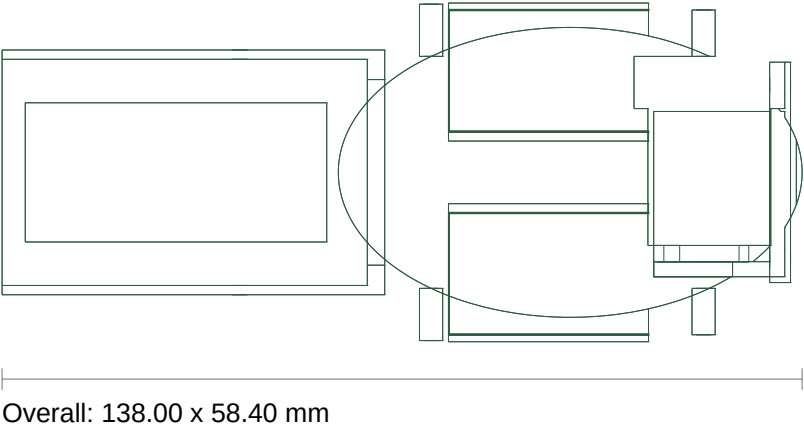
MICRO X Rev A / chassis

link: trunk | millimetres | original design, prototype reference, not production released

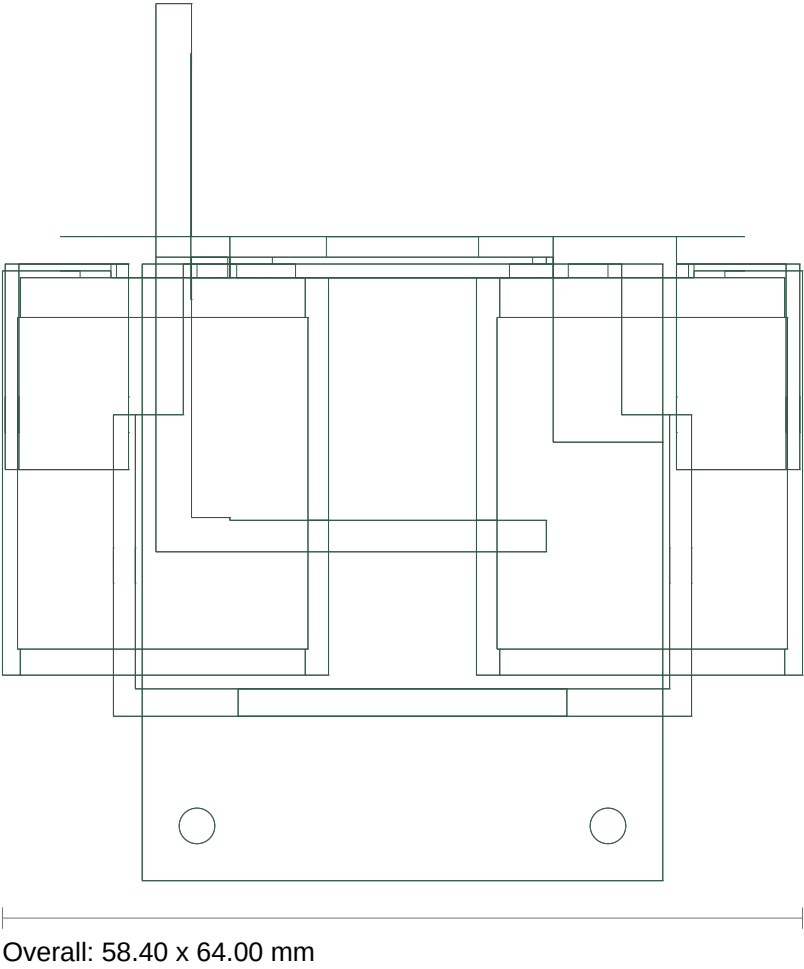
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 138.0 / 58.4 / 64.0 mm

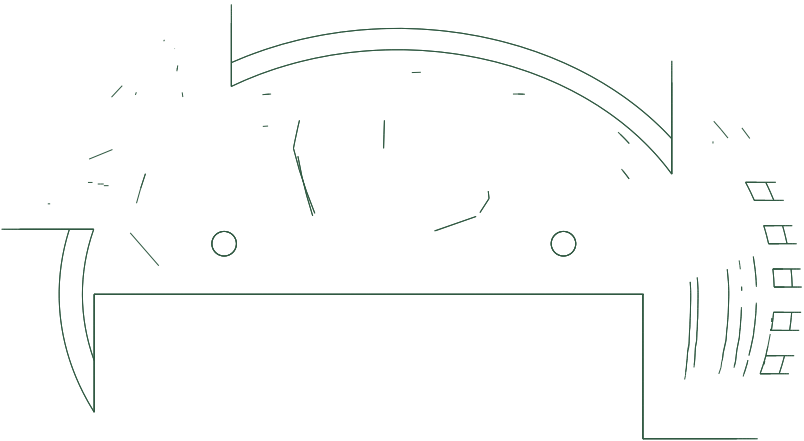
Material volume: 33749.0 mm3 | PLA solid-equivalent: 41.85 g (slicer mass will be lower)

Internal chassis: hip yaw hangers, neck servo bay, battery tray (NP-F550 class), chest board wall and four M3 shell bosses.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / torso_shell

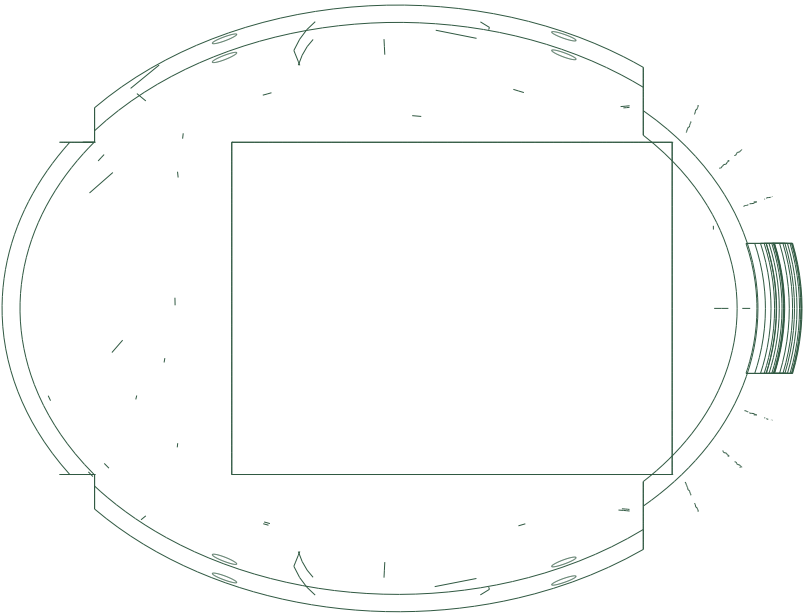
link: trunk | millimetres | original design, prototype reference, not production released

SIDE X-Z



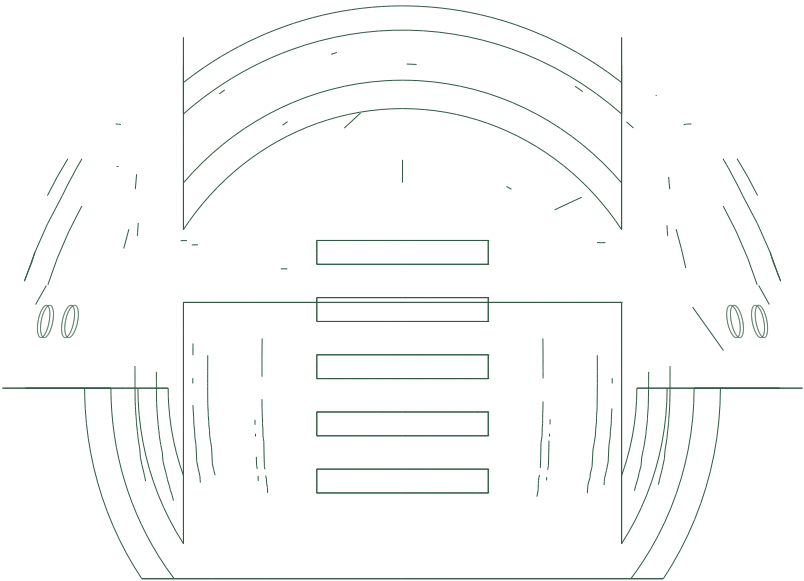
Overall: 110.81 x 60.12 mm

TOP X-Y



Overall: 110.81 x 83.99 mm

FRONT Y-Z



Overall: 83.99 x 60.12 mm

Assembly-frame extents X/Y/Z: 110.81 / 83.99 / 60.12 mm

Material volume: 26262.7 mm3 | PLA solid-equivalent: 32.57 g (slicer mass will be lower)

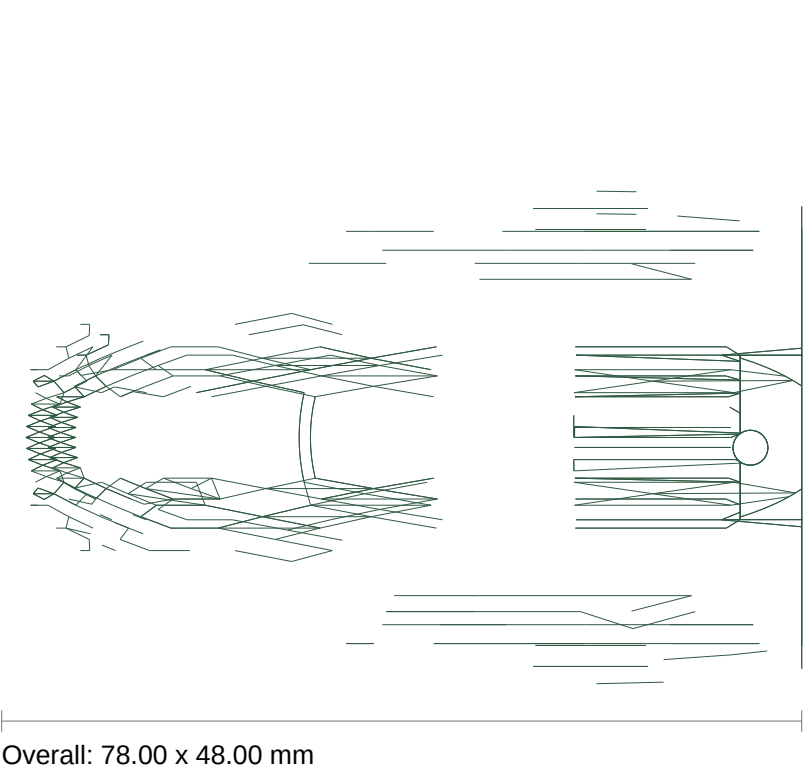
One-piece egg shell, open below the hips, with neck and tail-battery openings and a chest grille. Four M3 through the flanks into chassis bosses; remove it to reach the board.

Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

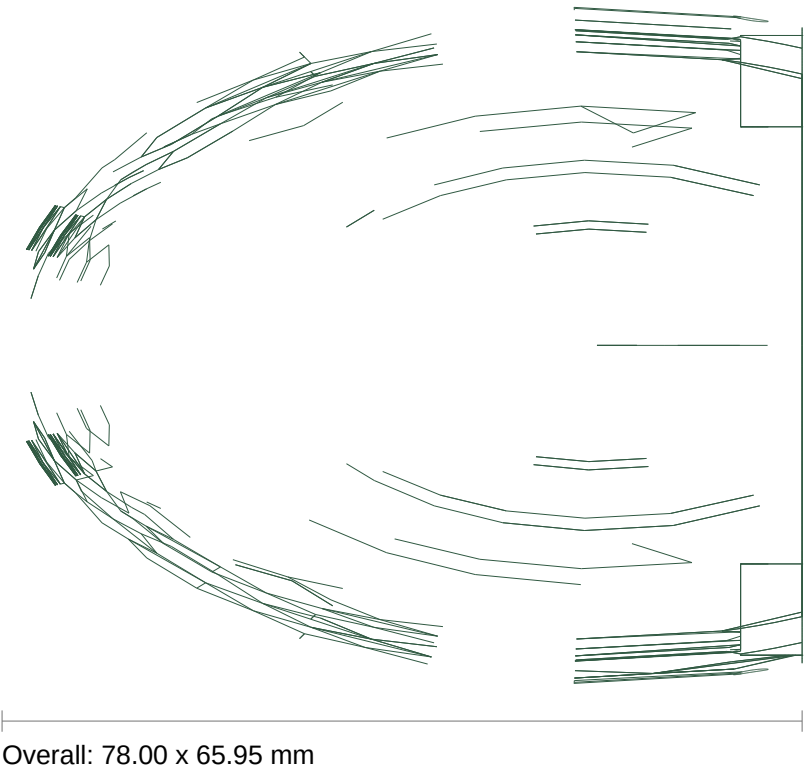
MICRO X Rev A / tail_cover

link: trunk | millimetres | original design, prototype reference, not production released

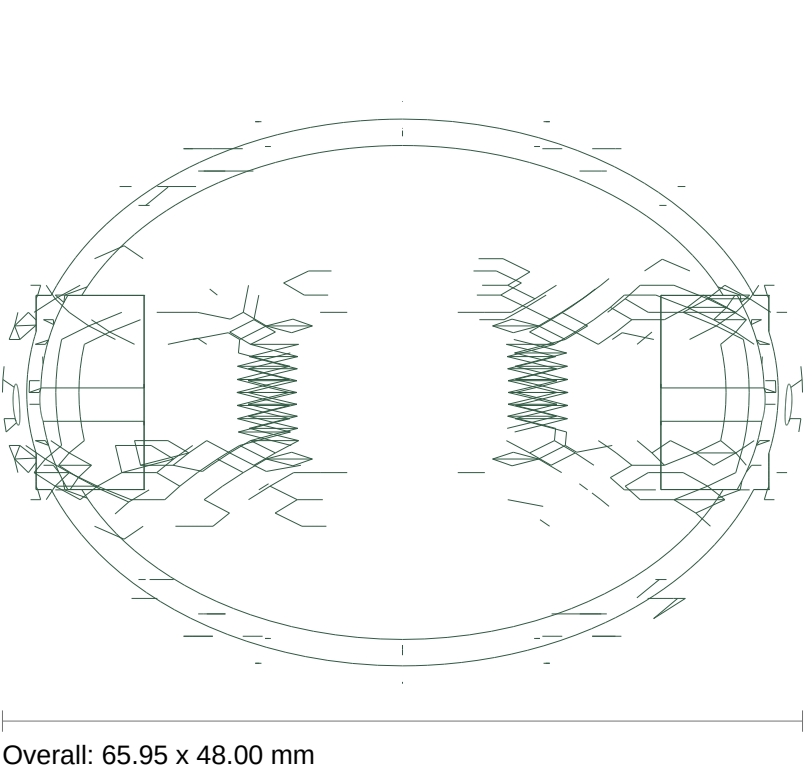
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 78.0 / 65.95 / 48.0 mm

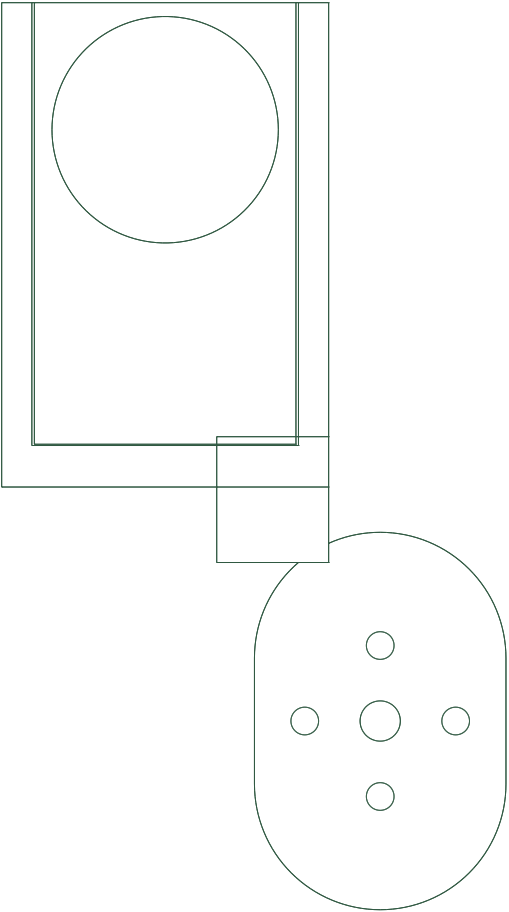
Material volume: 26391.1 mm3 | PLA solid-equivalent: 32.72 g (slicer mass will be lower)

Tail = battery cover: slides over the tray and holds an NP-F550 class pack; two M3 screws into the tray.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / neck_link

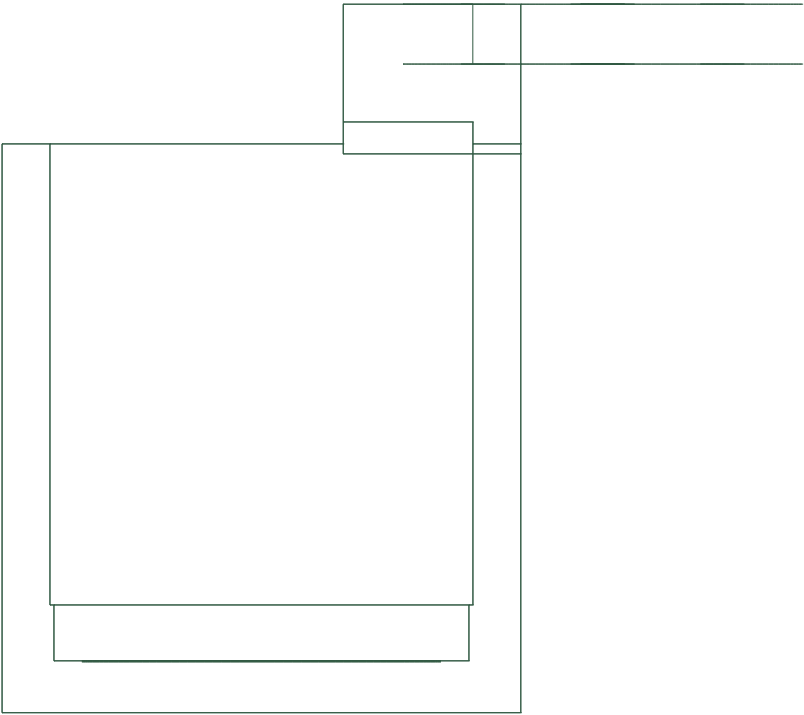
link: neck | millimetres | original design, prototype reference, not production released

SIDE X-Z



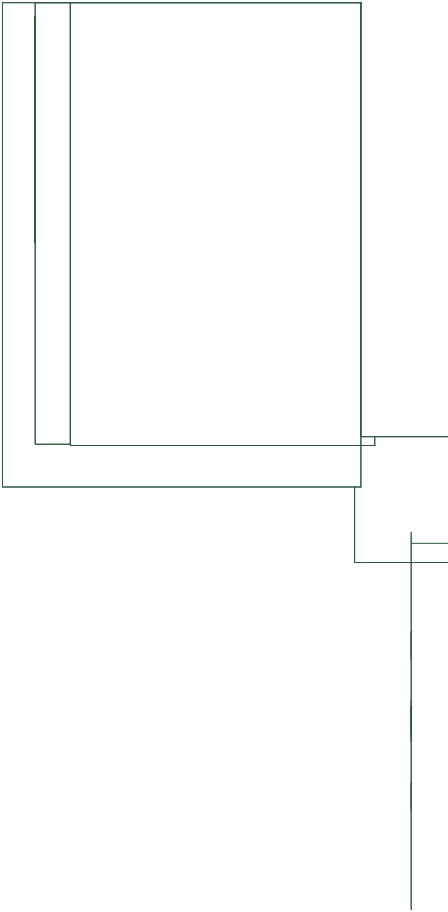
Overall: 40.10 x 72.10 mm

TOP X-Y



Overall: 40.10 x 35.50 mm

FRONT Y-Z



Overall: 35.50 x 72.10 mm

Assembly-frame extents X/Y/Z: 40.1 / 35.5 / 72.1 mm

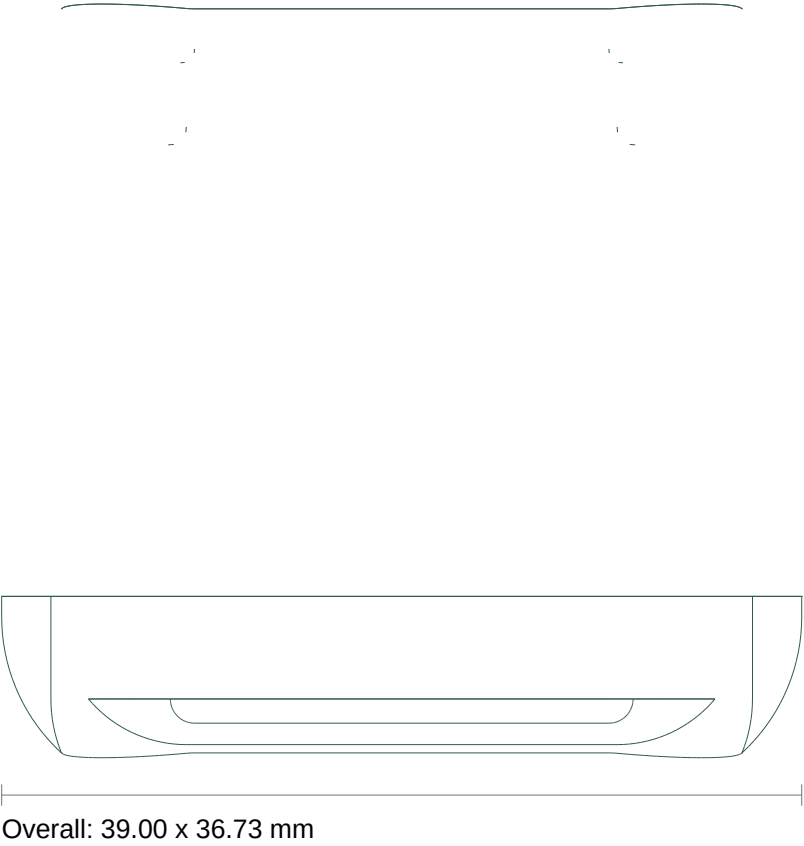
Material volume: 11345.3 mm3 | PLA solid-equivalent: 14.07 g (slicer mass will be lower)

Neck: neck-pitch horn plate rising into the head-pitch servo bay.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

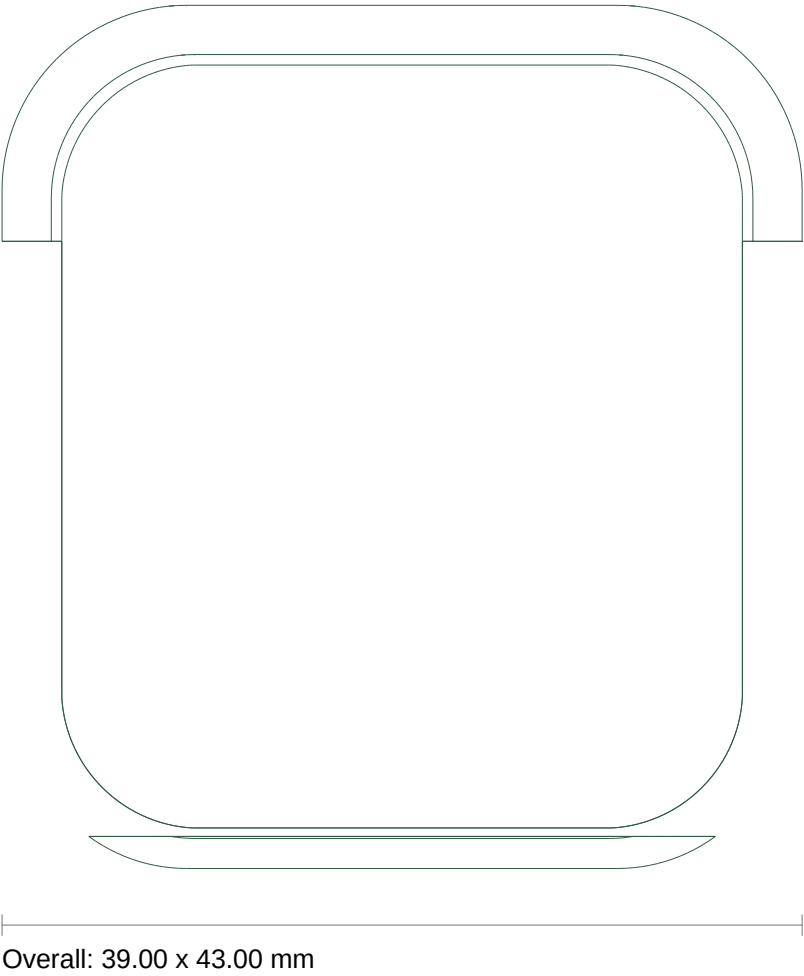
MICRO X Rev A / neck_sleeve

link: neck | millimetres | original design, prototype reference, not production released

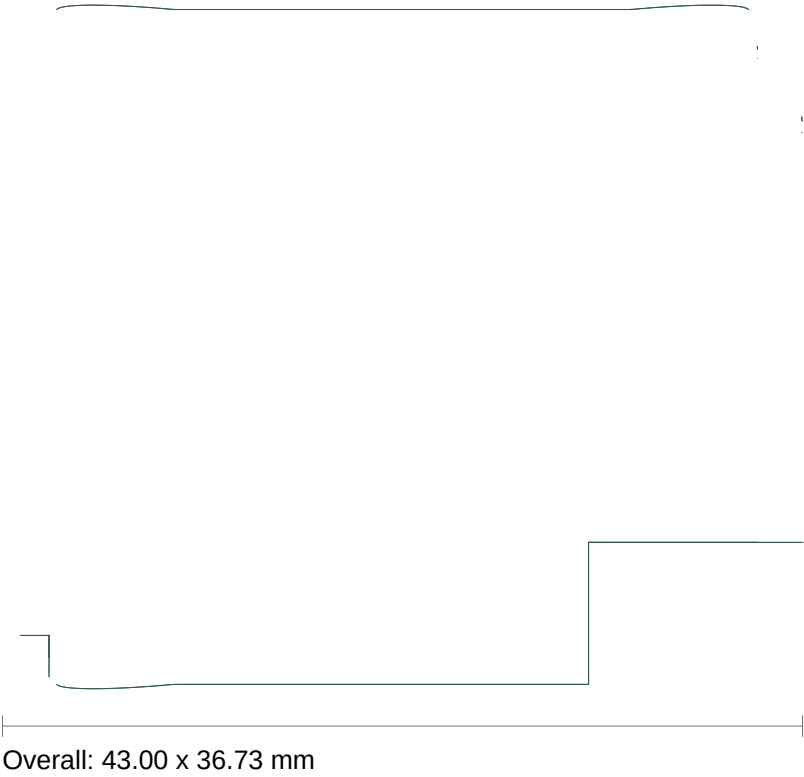
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 39.0 / 43.0 / 36.73 mm

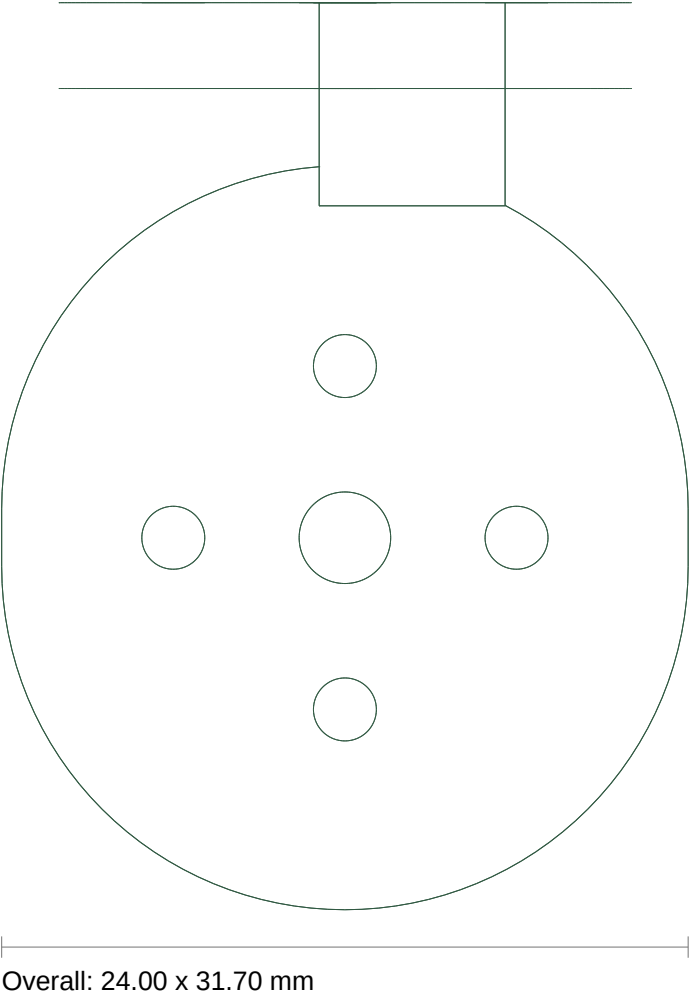
Material volume: 10147.4 mm3 | PLA solid-equivalent: 12.58 g (slicer mass will be lower)

Neck sleeve: rounded tube around the neck link and head-pitch servo; rotates with the neck inside the torso slot.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

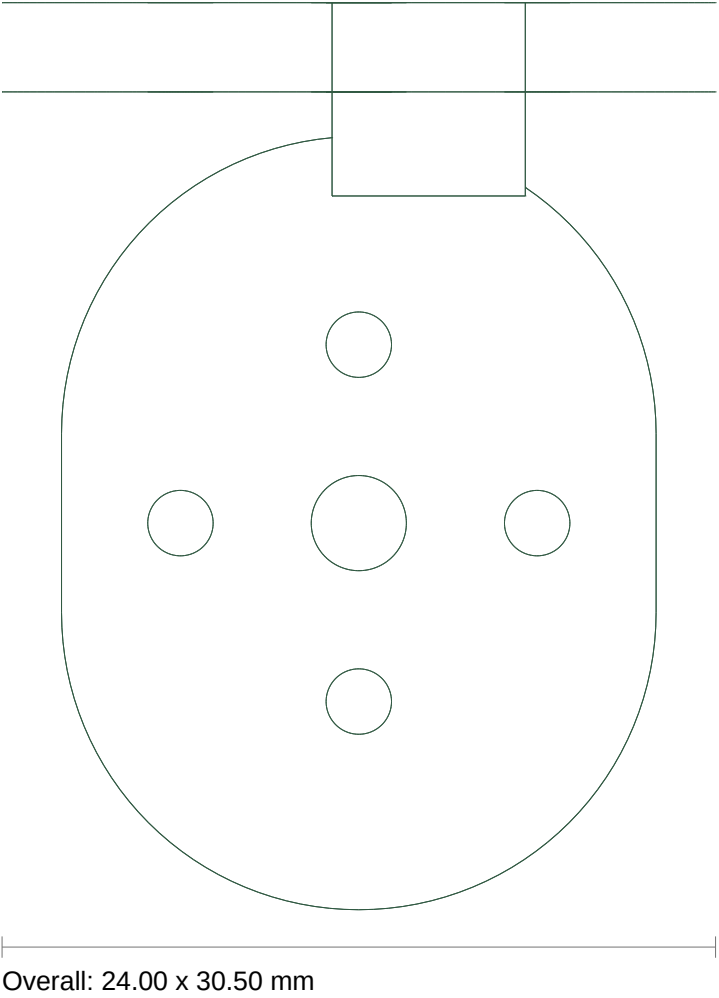
MICRO X Rev A / head_base

link: head_base | millimetres | original design, prototype reference, not production released

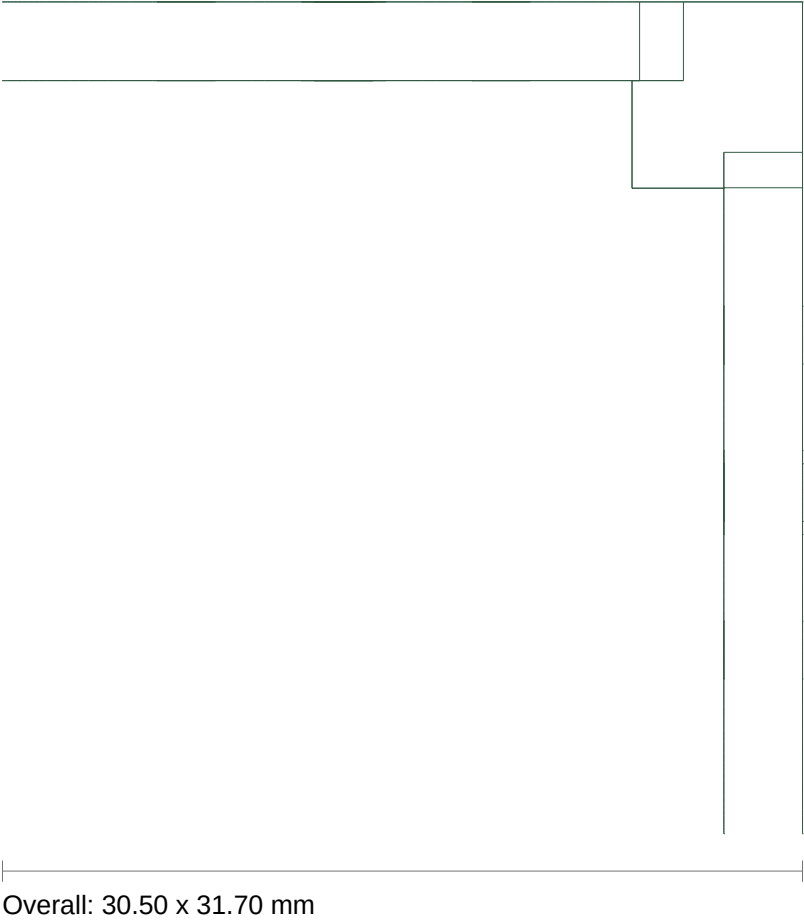
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 24.0 / 30.5 / 31.7 mm

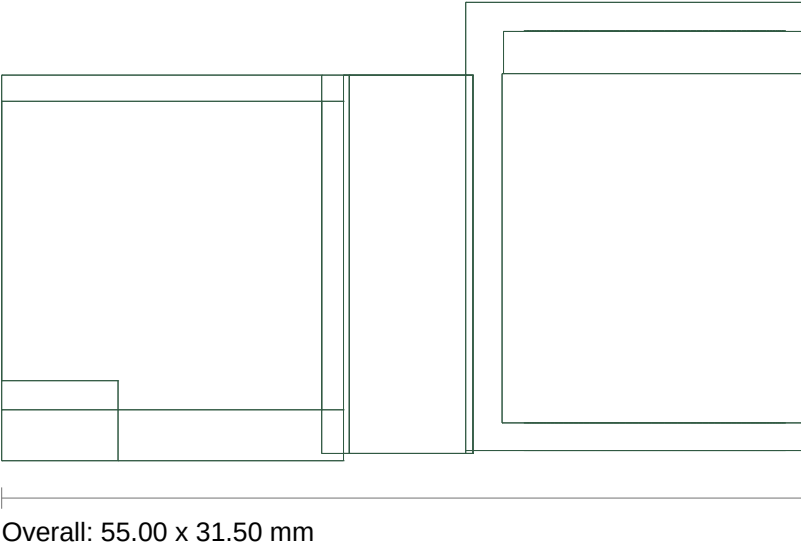
Material volume: 2915.8 mm3 | PLA solid-equivalent: 3.62 g (slicer mass will be lower)

Head base: head-pitch horn plate and the fixed head-yaw horn plate; the only part that stays still while the head yaws.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

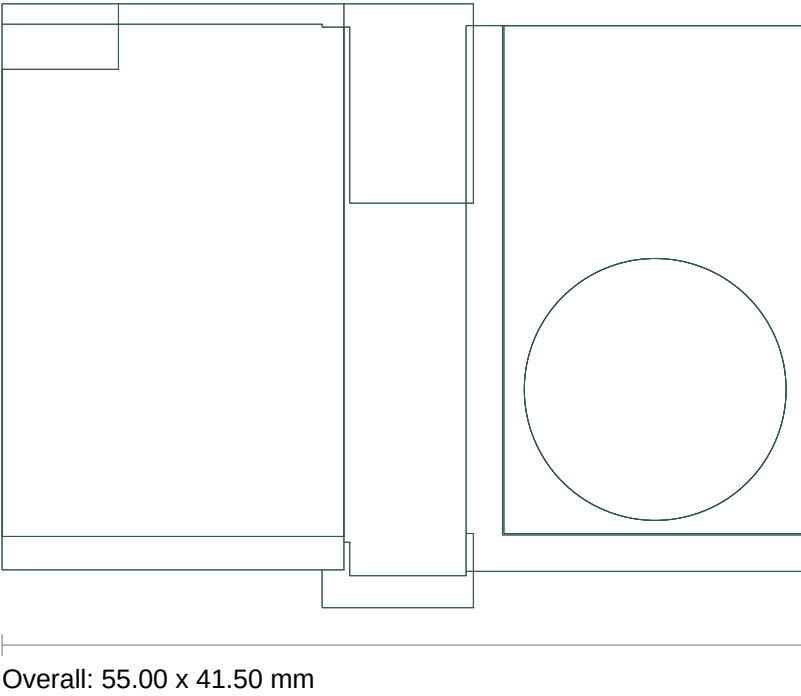
MICRO X Rev A / head_yoke

link: head_yoke | millimetres | original design, prototype reference, not production released

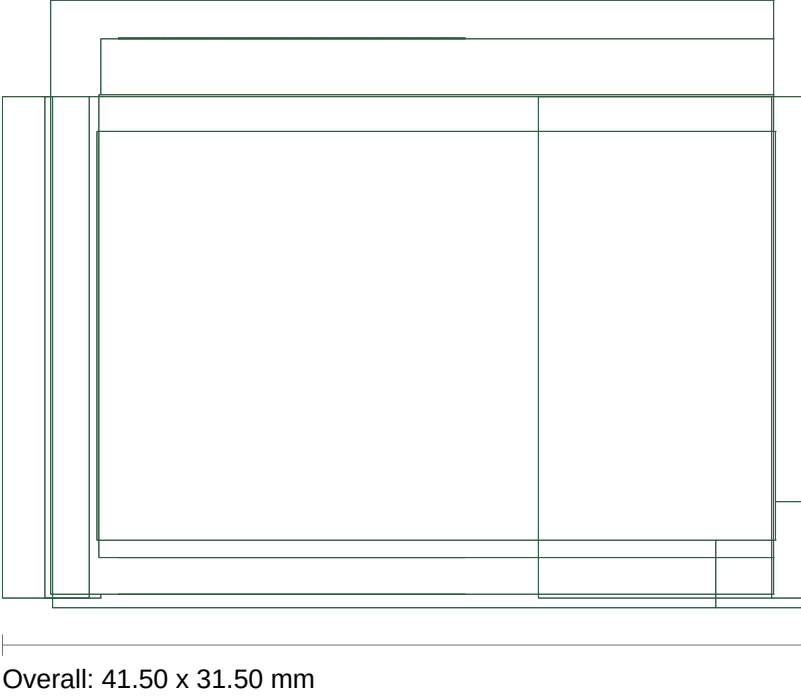
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 55.0 / 41.5 / 31.5 mm

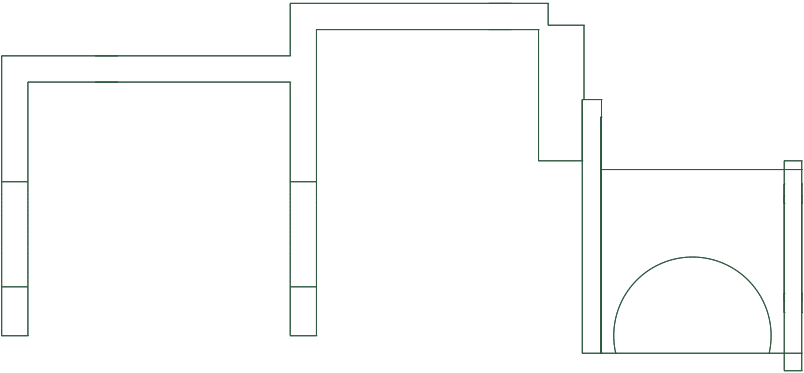
Material volume: 16877.3 mm3 | PLA solid-equivalent: 20.93 g (slicer mass will be lower)

Yoke: carries the head-yaw servo body (horn down onto the base) and the head-roll servo.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / head_frame

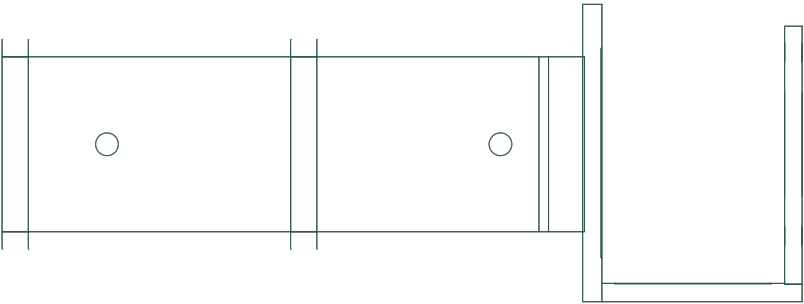
link: head | millimetres | original design, prototype reference, not production released

SIDE X-Z



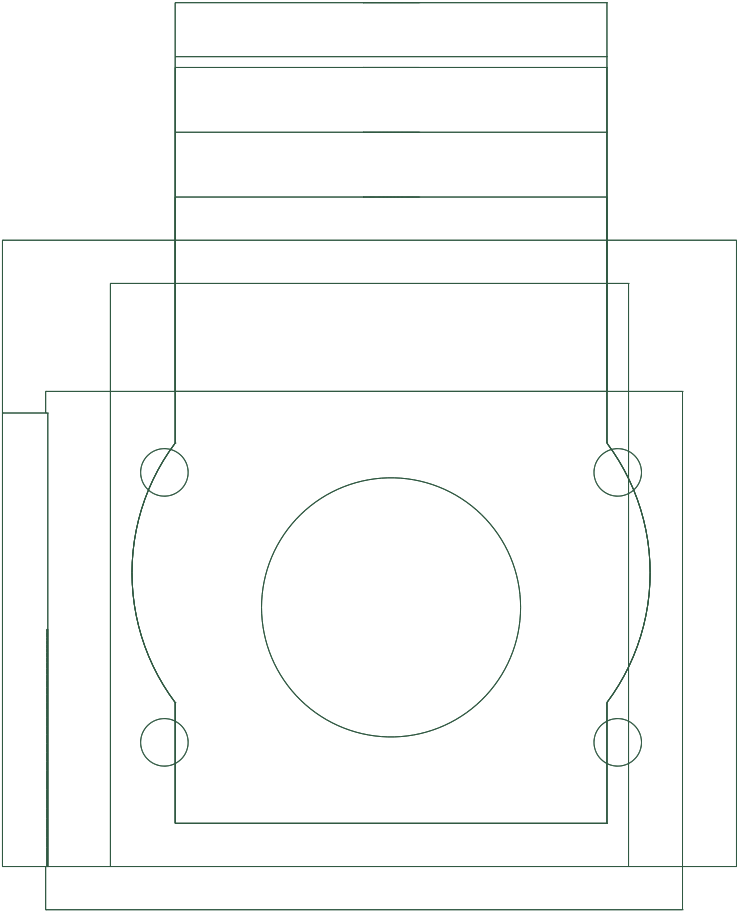
Overall: 91.50 x 42.00 mm

TOP X-Y



Overall: 91.50 x 34.00 mm

FRONT Y-Z



Overall: 34.00 x 42.00 mm

Assembly-frame extents X/Y/Z: 91.5 / 34.0 / 42.0 mm

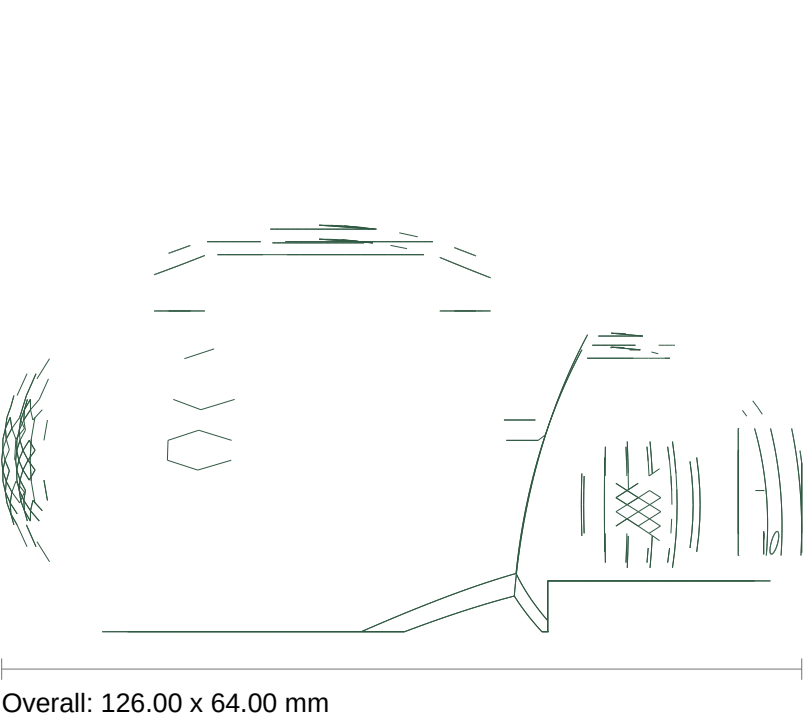
Material volume: 13589.1 mm3 | PLA solid-equivalent: 16.85 g (slicer mass will be lower)

Head frame: head-roll clevis, top rail, jaw servo bay and the camera carrier wall (Camera Module 3 hole pattern). Skull screws to the rail.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

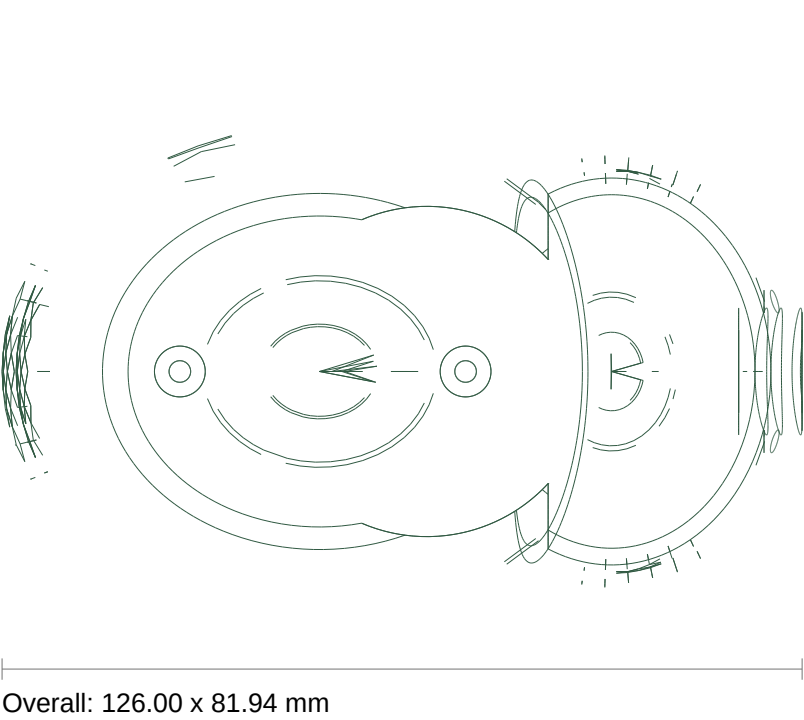
MICRO X Rev A / skull

link: head | millimetres | original design, prototype reference, not production released

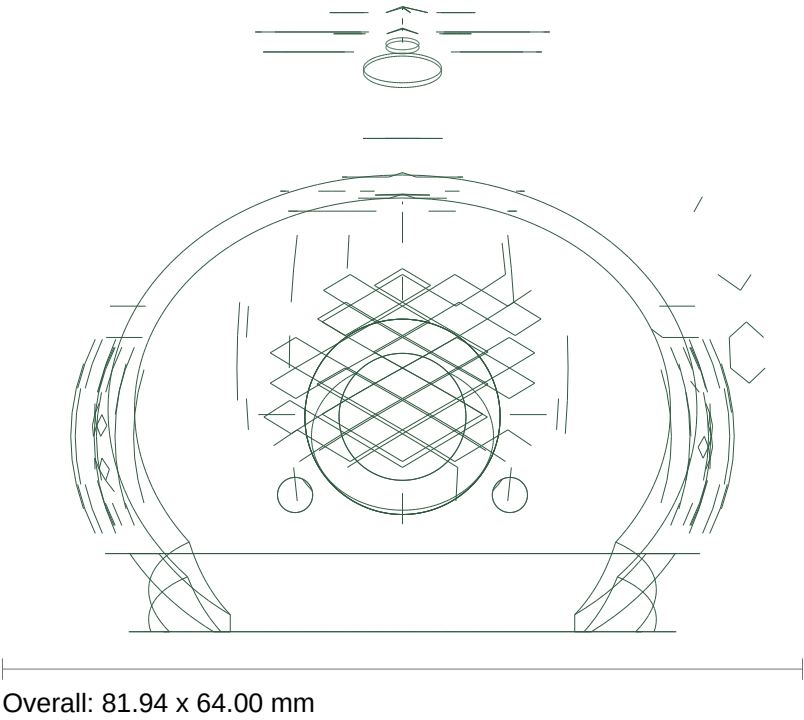
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 126.0 / 81.94 / 64.0 mm

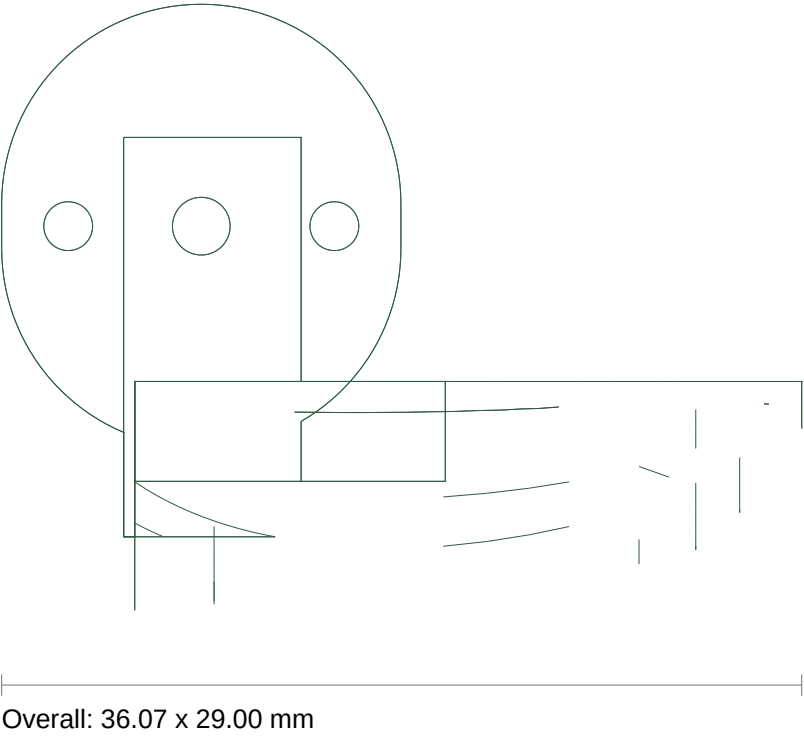
Material volume: 46630.3 mm3 | PLA solid-equivalent: 57.82 g (slicer mass will be lower)

Skull: one-piece hollow head with muzzle, camera ring and nostrils; no separate eye parts. Two M3 screws down into the head frame rail.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

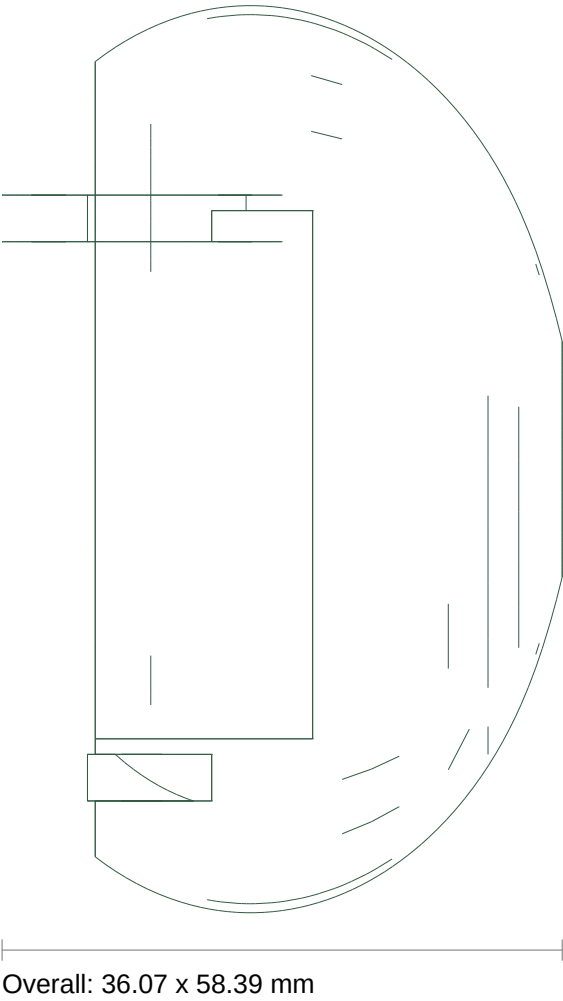
MICRO X Rev A / jaw_beak

link: jaw | millimetres | original design, prototype reference, not production released

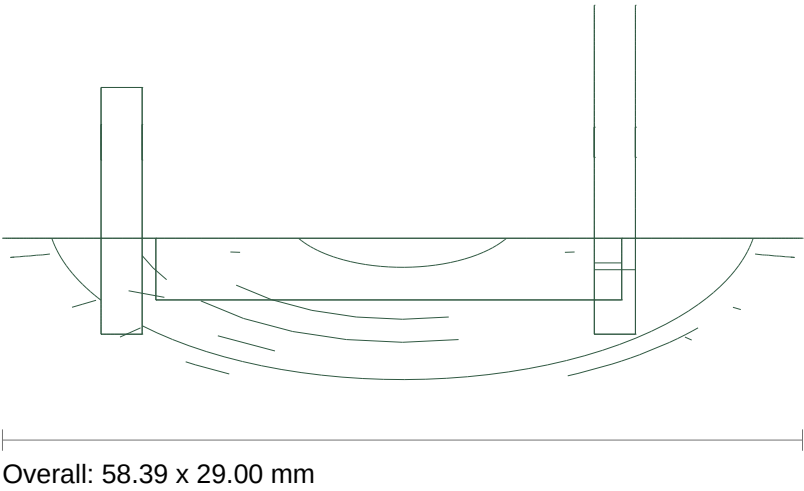
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 36.07 / 58.39 / 29.0 mm

Material volume: 10984.1 mm3 | PLA solid-equivalent: 13.62 g (slicer mass will be lower)

Lower beak: solid rounded scoop with a left arm bolted to the jaw servo horn and a right pin pivot; rear trimmed to clear the neck stack when the head yaws.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.